Estimating Unit Costs of Nutrients from Market Prices of Feedstuffs

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ABSTRACT

The determinations of nutrient costs and breakeven prices of feedstuffs are important activities in the field of nutritional economics. Current methods are either extensions of the Petersen method, which dates back to 1932, or mathematical programming methods generally set as linear programming problems. Both sets of methods have severe limitations that make them unsuitable or biased for deriving unit costs of nutrients and aggregate breakeven costs of feedstuffs in a given market. We propose a least-squares method that, under verifiable conditions, yields maximum likelihood estimates of unit costs of nutrients and breakeven prices of feedstuffs. Assumptions to the maximum likelihood estimates are verified with studentized residuals, leverage values, and variance inflation factors. The method has been programmed into a Windows application and can be used by those with little formal training in statistics.
(**Key words:** nutrient costs, breakeven prices, Peterson method, maximum likelihood estimates)

Abbreviation key: BAR = ground barley, **CAM** = canola meal, **EBM** = equation-based methods, **GSC** = ground shelled corn, **IBM** = inequation-based methods, **LP** = linear programming, **ML** = maximum likelihood, **MT** = metric tonne, **PM** = Peterson method, **SBM** = dehulled, solvent-extracted soybean meal, **VIF** = variance inflation factors.

INTRODUCTION

A variety of methods have been proposed to estimate unit costs of nutrients and, implicitly or explicitly, the feeding value of feedstuffs. All methods fall into one of two general categories: equation-based (**EBM**) and inequation-based methods (**IBM**). For EBM, a set of equations developed from the nutritional composition of referee feeds is solved by using their market prices. The best-known method among this group is the Petersen Method (**PM**) in which the energy and protein compositions of corn grain and soybean meal are equated to their respective prices, setting a set of two equations with two unknowns. The method dates back to 1932 (14) and is presented and discussed at length by Morrison (11). Although widely used, the method is fundamentally flawed in that it assumes efficient markets in commodity trading and implies economically incoherent behavioral patterns by buyers and sellers of commodities.

The second series of methods, IBM, are basically constrained optimization models solved using mathematical programming techniques (1, 20). Linear programming (**LP**) is the best known member of this group and became widely used in animal nutrition with the discovery of an efficient algorithm (3) and the advent of high-speed computers. Within an LP model, a cost function is minimized subject to a series of inequations forcing the solution to meet the nutritional requirements of the animal for which the diet is being optimized. Linear programs suffer from being case specific, and they deliver little information on the unit costs of nutrients. Nutrients with nonbinding constraints have an implicit unit cost of zero. Shadow costs of binding nutrients provide information on unit costs that are valid only at the margin. Additionally, the information delivered has a very narrow inference range because it provides estimates that are sound only for one group of animals in a given herd. Consequently, LP is limited in providing estimates of aggregate unit costs of nutrients within a given market. This research attempts to circumvent these problems by developing a new procedure to provide estimates of aggregate unit costs of nutrients and breakeven prices of feedstuffs based on the trading of all feed commodities in a given market.

PROCEDURE DERIVATION

The Petersen Method

In the PM method (14), prices of n feedstuffs are equated to their composition in n nutrients. In most

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published applications, n has been set to 2 or 3. In the United States, TDN and CP have traditionally been used, although more recent applications have included other nutrients. In nearly all cases, US applications have used corn and soybean meal as the referee feeds. Without loss of generality, we can setup our equations using NE_L and CP for nutrients and ground, shelled corn (**GSC**) and dehulled, solvent extracted (48%) soybean meal (**SBM**) as the referee feeds. According to NRC (13), GSC and SBM contain 1.96 and 2.01 Mcal/kg NE_L, and 100.0 and 551.0 g/kg of CP on a DM basis, with a moisture content of 120 and 100 g/kg for GSC and SBM, respectively. Therefore, a metric tonne (**MT**) of GSC contains 1724.8 Mcal of NE_L and 88.0 kg of CP on an as-fed basis. Likewise, one MT of SBM contains 1809.0 Mcal of NE_L and 495.9 kg of CP. In the PM scheme, the following system of two equations with two unknowns is set:

$$\begin{aligned} GSC\$ &= 1724.8\ NE\$ + 88.0\ CP\$ \\ SBM\$ &= 1809.0\ NE\$ + 495.9\ CP\$ \end{aligned} \qquad [1]$$

Where

GSC$ = cost of GSC per MT,
SBM$ = cost of SBM per MT,
NE$ = cost per Mcal of NE_L (unknown), and
CP$ = cost per kg of CP (unknown).

This system of equations is easily solved. As an example, NE$ = 0.046 and CP$ = 0.236 when GSC$ and SBM$ are set at $100 and $200 per MT, respectively. These estimates of NE$ and CP$ can then be used to calculate the implicit values, often referred to as breakeven prices, of all other commodities in a given market.

Adding a nutrient to the estimation process is straightforward algebraically. An additional referee feed must be used for each additional nutrient. The software Feedval 4 from the University of Wisconsin, for example, sets six equations to solve for six nutrients. (Feedval 4 is available as a Microsoft Excel™ spreadsheet via the worldwide web at www.wisc.edu/dysci/vwex/nutritn/spreadshees/sprds.htm.) Referee feeds are blood meal, urea, corn grain, tallow, limestone, and dicalcium phosphate and are used to estimate the unit costs of Ca, P, RUP, RDP, fat, and TDN.

Although useful, the PM method contains fundamental flaws that cannot be ignored. First, referee feeds are never either well or poorly priced. That is, the method implicitly assumes that referee feeds are always priced at their breakeven prices. Invariably, one finds other commodities that are priced under their own breakeven prices. Hence, one would conclude that referee feeds should never be purchased based on a PM evaluation. This is an odd conclusion because the method implicitly assumes that referee feeds are market movers and set the prices of other commodities in the marketplace. The second flaw of the method is that PM assumes perfectly competitive markets. This implies that those trading referee feeds have perfect market information, with trading occurring at a perfect equilibrium point between supply and demand. Indirectly, this implies that the dairy industry must have a significant effect on the total demand of referee feeds (GSC and SBM in our example), a conclusion that is not supported by the current relative demand by the dairy industry for most feedstuffs. The dairy industry accounts for less than 10% of the total US demand for corn and SBM and for much less of the total demand for any of the reference feeds used by Feedval 4 (17). Third, the application of PM over a long period of time (years) results in an implicit incoherent economic behavior by buyers and sellers. That is, buyers keep purchasing some commodities well above their breakeven prices while sellers keep selling other commodities at prices considerably less than their breakeven prices. Fourth, it is difficult to augment PM to accommodate additional nutrients, not because of the algebra involved, but because of the difficulty in identifying proper referee feeds. With more than three or four nutrients involved, the system of equation tends to become ill-conditioned (4), with the consequence that a small change in the price of one referee feed results in substantial changes in the estimated costs of nutrients. This ill-conditioning characteristic of the equation system also applies to the nutritional composition coefficients. That is, one must assume near-perfect knowledge of the composition of referee feeds. This assumption may be reasonable for CP, a poor indicator of biological value in ruminants, but is greatly challenged when nutrients with more uncertain characteristics, such as RUP and NE_L are being considered.

Multiple Petersen Method

Some of the problems associated with the PM can be alleviated by considering more than two feedstuffs for the estimation of the unit cost of two nutrients. To understand how this can be accomplished, it helps to recast [1] using a more general notation as follows:

$$[2]\ Y_i = \sum_{j}^{2} X_{ij}B_j \quad i = 1,2$$

where

Y_i = price per MT of feedstuffs i,

X_{ij} = amount of nutrient j in a MT of feedstuffs i, and
B_j = cost per unit of nutrient j.

The equation set in [2] can be solved for any two feedstuffs (we will ignore here the obvious difficulty resulting from two feedstuffs of identical composition or, more generally, from collinearity between two equations i and i′ with $i \neq i'$). For example, we could evaluate the unit costs of NE_L and CP using GSC, SBM, ground barley (**BAR**), and canola meal (**CAM**), and use the standard nutritional composition reported by NRC (13), and price per MT of \$90, \$150, \$100, and \$200, respectively, for BAR, CAM, GSC and SBM. If we use BAR and CAM to set the first set of equations, we get:

$$\text{BAR:} \quad 90 = 1707.2\ \text{NE\$} + 118.8\ \text{CP\$} \qquad [3]$$

$$\text{CAM:} \quad 150 = 1582.4\ \text{NE\$} + 404.8\ \text{CP\$}$$

Likewise, a second set of equations can be set using BAR and GSC:

$$\text{BAR:} \quad 90 = 1707.2\ \text{NE\$} + 118.8\ \text{CP\$} \qquad [4]$$

$$\text{GSC:} \quad 100 = 1724.8\ \text{NE\$} + 88.0\ \text{CP\$}$$

With four feedstuffs, we can set six systems of paired equations and get six separate, although not independent, estimates of \$NE and \$CP. These estimates are reported in Table 1. Clearly, the selected pair of feedstuffs has a dramatic effect on the estimates. Across all pairs, mean estimates have SE that are approaching the magnitude of the means. Also, NE\$ and CP\$ estimates are highly correlated: the Pearson product-moment correlation (18) equals –0.85 ($P < 0.03$) and the Spearman rank correlation (18) equals –0.94 ($P < 0.005$).

Table 1. Estimated unit cost of NE_L and CP using the Multiple Petersen Method with ground barley (BAR), canola meal (CAM), ground shelled corn (GSC), and dehulled, solvent extracted soybean meal (SBM)[1].

Pair of Feedstuffs	NE_L (\$/Mcal)	CP (\$/kg)
BAR, CAM	0.037	0.226
BAR, GSC	0.724	–0.283
BAR, SBM	0.033	0.283
CAM, GSC	0.049	0.180
CAM, SBM	–0.125	0.861
GSC, SBM	0.046	0.236
Mean estimate	0.127	0.251
SEM	0.122	0.149
Median estimate	0.042	0.231

[1]Using the following cost per metric ton: BAR = \$90, CAM = \$150, GSC = \$100, and SBM = \$200.

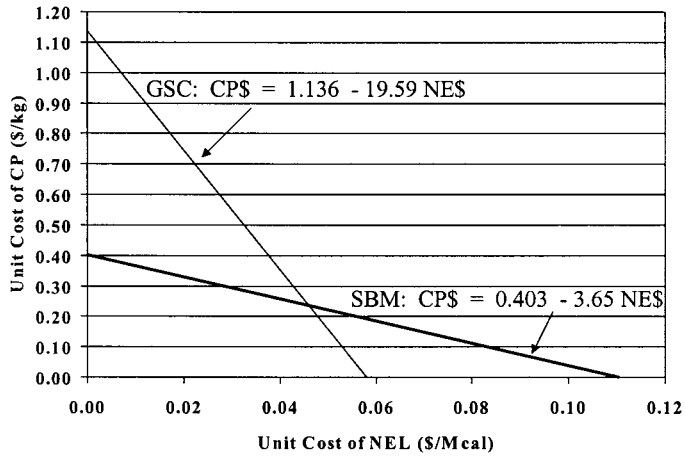

Figure 1. Graphical presentation of the relatively well-conditioned system of equations for ground, shelled corn (GSC) and dehulled, solvent extracted soybean meal (SBM) where unit cost of CP (CP\$) and unit cost of NE_L (NE\$) are estimated.

This method of estimation is fundamentally flawed because it assumes that all systems of paired equations have the same accuracy of estimation. This assumption ignores the fact that the condition number, defined as the ratio of the maximum to the minimum stretch induced by the X_{ij} coefficients, varies dramatically across systems of equations (4). For a system of two equations, the condition number can be easily presented in graphical form. Figure 1 shows the system of equations for GSC and SBM. Although not orthogonal, the two lines have slopes that are sufficiently different (GSC = –19.58, SBM = –3.65) for a relatively accurate estimate of their point of intersection. This result is in contrast with the system of equations of CAM and SBM presented in Figure 2. The graph of the two equations shows that they are nearly parallel lines (slope of CAM = –3.91), so that moving one of the two lines a little would drastically alter their point of intersection. This last system of equations has a high condition number and the system is ill-conditioned, meaning that the solution is very sensitive to small changes in the data (composition or price).

Although the multiple Petersen method appears ill-suited for the task of estimating unit costs of nutrients, we chose to present it because it bridges the concepts behind PM and the maximum likelihood method (**ML**) that we have developed.

Maximum Likelihood Method

Two important changes must be made in our conceptual approach to the problem. First, we must move away from the concept of a few referee feedstuffs. We shall instead use an approach in which all feedstuffs are

used jointly and simultaneously to estimate unit costs of nutrients. From a market point of view, this implies that no feedstuffs rule the market. Second, we will no longer assume efficient markets. That is, we will implicitly recognize that markets are not instantaneously self-correcting. At any point in time, any feedstuff can be priced over or under its breakeven price. This is accomplished by adding an error term to equation [2] as follows:

$$Y_i = \sum_{j}^{2} X_{ij}B_j + e_i \qquad [5]$$

This equation can be easily expanded to more than two nutrients:

$$\mathbf{Y} = \mathbf{XB} + E \qquad [6]$$

where

Y is an n × 1 vector of prices for n feedstuffs,
X is an n × m matrix of coefficients of m nutrients for n feedstuffs,
B is an m × 1 vector of unit costs of m nutrients, and
E is an n × 1 vector of errors.

The system of equations [6] has an infinite number of solutions because it contains more unknowns than the number of equations (n unknown e_i and m unknown B_j in a system of n equations). The system, however, has a unique solution if we set for objectives to minimize the sum of squares of deviations (i.e., minimize $\sum_i e_i^2$). In matrix notation, this implies the minimization of:

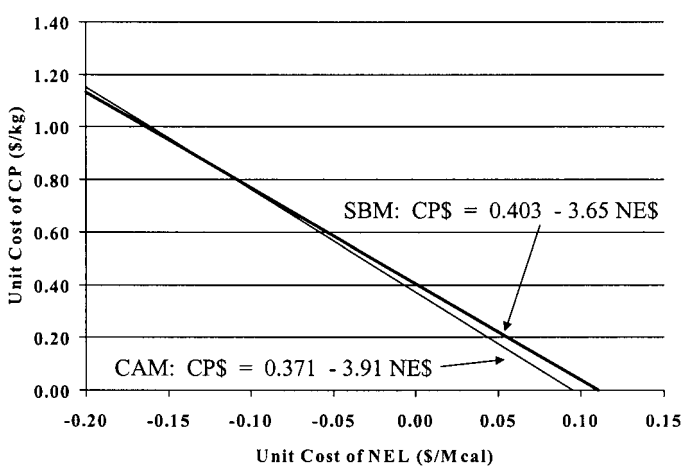

Figure 2. Graphical representation of the ill-conditioned system of equations for canola meal (CAM) and dehulled, solvent extracted soybean meal (SBM) where unit cost of CP (CP$) and unit cost of NE_L (NE$) are estimated.

$$\mathbf{E'E} = \mathbf{Y'Y} - 2\mathbf{B'X'Y} + \mathbf{B'X'XB} \qquad [7]$$

Differentiating [7] with respect to **B** and setting the resulting matrix equation equal to a vector of zeros, the least-squares estimate of **B** is the vector **b**:

$$\mathbf{b} = (\mathbf{X'X})^{-1}\mathbf{X'Y} \qquad [8]$$

This solution **b** has a unique set of properties. First, **b** is an estimate of **B** that minimizes the error sum of squares **E′E** regardless of any distribution properties of the errors. Second, the elements of **b** are linear functions of Y_1, Y_2, …, Y_n and provide unbiased estimates of **B**. These estimates have the minimum variances among any estimates that are linear functions of the Y's, irrespective of distribution properties of the errors. Last, and importantly, if the errors are independent and identically distributed from a normal distribution with a mean of 0 and a variance of σ^2, i.e., $E \approx N(\mathbf{O}, \mathbf{I}\sigma^2)$, then **b** is the ML estimate of **B**. Therefore, equation [8] produces ML estimates of the unit costs of nutrients using market prices and composition of feedstuffs if the residuals are independent, have common variance and are all distributed according to a normal distribution with a mean of zero and a variance of σ^2. Consequently, it will be important in the application of our ML method to ensure that these assumptions are realistic.

A direct benefit of using a ML approach is that various statistics can be calculated using conventional, well-known statistical methods (5). In particular, an estimate of the variance (σ^2) is:

$$\begin{aligned}\sigma^2 &= \mathbf{Y'Y} - \mathbf{Y'X}\,(\mathbf{X'X})^{-1}\mathbf{X'Y}/n - m\\ &= \mathbf{Y'Y} - \mathbf{Y'Xb}/n - m\\ &= \mathbf{Y'Y} - \mathbf{b'X'Y}/n - m\end{aligned} \qquad [9]$$

The variance of estimated unit costs of nutrients **b** is:

$$\mathrm{Var}(\mathbf{b}) = (\mathbf{X'X})^{-1}\sigma^2 \qquad [10]$$

Lastly, the variance of the predicted breakeven price of a feedstuffs is given by:

$$\mathrm{Var}(\hat{Y}_i) = \mathbf{X_i}(\mathbf{X'X})^{-1}\mathbf{X_i}\sigma^2 \qquad [11]$$

Figure 3 reports the results obtained when using the ML method on our simple example consisting of BAR, CAM, GSC, and SBM. Although purposely kept simple, this example illustrates some of the appealing and unique features of a ML approach: estimates of unit costs of nutrients have standard errors, correlations between estimates is explicit, and range as opposed to point estimates of breakeven prices of feedstuffs can be calculated relatively easily.

$$X = \begin{bmatrix} 1707.2 & 118.8 \\ 1582.4 & 404.8 \\ 1724.8 & 88.0 \\ 1809.0 & 495.9 \end{bmatrix} \qquad Y = \begin{bmatrix} 90 \\ 150 \\ 100 \\ 200 \end{bmatrix}$$

$$X'X = \begin{bmatrix} 1.16660E7 & 1.8922E6 \\ 1.18922E6 & 4.3164E5 \end{bmatrix} \qquad X'Y = \begin{bmatrix} 9.2529E5 \\ 1.7939E5 \end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix} 2.9668E-7 & -1.3006E-6 \\ -1.3006E-6 & 8.0184E-6 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.0412 \\ 0.2350 \end{bmatrix} \qquad VAR(b) = \begin{bmatrix} 4.7868E-5 & 2.0985E-4 \\ 2.0985E-4 & 1.2937E-3 \end{bmatrix}$$

$$r_{b_1, b_2} = 0.843$$

$$\hat{Y} = \begin{bmatrix} 98.25 \\ 160.32 \\ 91.74 \\ 191.07 \end{bmatrix} \qquad VAR(\hat{Y}) = \begin{bmatrix} 72.65 \\ 63.02 \\ 88.70 \\ 98.26 \end{bmatrix}$$

Figure 3. Matrices involved in the least-squares, maximum likelihood estimation of the unit cost of NE_L (NE$) and the unit cost of CP (CP$), using ground barley, canola meal, ground shelled corn, and dehulled, solvent extracted soybean meal. Composition used was from NRC (13). (Prices were set at $90, $150, $100, and $200 per metric ton for ground barley, canola meal, ground shelled corn, and soybean meal, respectively.)

ESTIMATION PROBLEMS

The Intercept Question

The inclusion of an intercept term in model [6] allows the estimation of an overall average residual, i.e., a number equal to the average value of feedstuffs after the value of the nutrients have been accounted for. In general, this term should be close to zero. Failure to include an intercept term when indeed one is present in the population of feedstuffs being studied leads to biased estimates of the other elements of **b**. In our application (19), we recommend to first use a model that includes an intercept, to test for the significance of such intercept, and to remove the intercept term when it is nonsignificant before reestimating **b**.

Outliers

The term outlier refers to an observation that is extreme relative to the bulk of the remaining observations. The presence of one or more outliers invalidates the assumption of normal distribution of errors in [6], resulting in an estimate of **b** that is no longer ML. Many different rules have been proposed for identifying and rejecting outliers (2, 15, 16). However, the automatic rejection of outliers is not always a very wise procedure. At times, the outlier is giving information not provided by other observations. In our application (19), we chose not to include an automatic rejection scheme of possible outliers. The user is provided with studentized residual measures, and observations with studentized residuals

that exceed ± 2.5 standard deviations should be looked at closely to ensure that the nutritional content of the feedstuffs is realistic and that the market price used is real.

Leveraged Observations

The detection of leveraged observations is a topic that receives little coverage, although leveraged observations can result in substantial problems in parameter estimation (10). A leverage point is an outlier among the predictor variables. Leverage points can be identified statistically from the leverage values h_i, which are the diagonal elements of the projection matrix (8):

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad [12]$$

Thus, we can rewrite [11] as follows:

$$\mathrm{Var}(\hat{Y}_i) = h_i\sigma^2 \quad [13]$$

Expressed this way, the effect of the leverage value on the variance of the estimate of the breakeven price for feedstuffs i is clearly seen.

In our application (19), feedstuffs with leverage values $h_i > 2\ (p + 1)/n$, where p = number of parameters and n = number of feedstuffs, are deemed leveraged and their inclusion in the data set should be questioned. Feedstuffs with leveraged values $h_i > 3\ (p + 1)/n$ should almost invariably be removed from the dataset.

Collinearity

Collinearities are approximate linear dependencies among the columns of predictor variable values. Collinearities can be due to model constraints, population characteristics that restrict variate values, sampling deficiencies, or overdefined models. Collinearities produce large coefficient estimates and inflate the variance of parameter estimates (9). In short, collinearity induces ill-conditioning of the $\mathbf{X}'\mathbf{X}$ matrix, that is, the determinant of $\mathbf{X}'\mathbf{X}$ is close to zero. This condition can be thought of in two ways: either, the model is overspecified (we are trying to estimate the cost of more nutrients than are needed to express the breakeven prices of the feedstuffs selected) or the data are not adequate to estimate all of the selected nutrients.

Collinearities can be measured with variance inflation factors (**VIF**) which are the diagonal elements of $(\mathbf{X}'\mathbf{X})^{-1}$ when $\mathbf{X}'\mathbf{X}$ is scaled to correlation form. In general, estimability is acceptable when all VIF < 10 (by definition, VIF = 1 in the absence of collinearity). In instances where VIF > 20, action must be taken. Additional feedstuffs with different nutritional characteristics can be included in the data, and this may result in

Table 2. Composition of feedstuffs and Midwest market prices in June 1999 used for the estimation of the unit cost of NDF, RUP, RDP, and NE_L[1].

	NDF (%)	RUP (%)	RDP (%)	NE_L (Mcal/kg)	Cost ($/metric tonne)	Location
Barley, grain	16.720	3.929	7.951	1.705	87.50	Kansas City
Blood meal	0.828	66.387	14.573	1.377	345.00	Minneapolis
Brewers grains, dehydrated	42.320	15.395	7.973	1.377	72.00	Chicago
Corn gluten meal, 60% CP	12.600	35.683	24.797	1.854	210.00	Chicago
Corn gluten feed	40.500	5.760	17.280	1.722	55.00	Chicago
Corn grain, ground	7.920	4.869	3.931	1.727	78.57	Chicago
Soybean meal, solvent-extracted, 44% CP	13.261	15.544	28.867	1.725	125.00	Chicago
Beet pulp, dehydrated	49.140	5.064	3.763	1.630	125.00	Chicago
Molasses, sugarcane	0.000	0.000	4.350	1.233	65.00	Minneapolis
Canola meal, solvent-extracted	24.752	11.878	25.068	1.429	96.00	Minneapolis
Cottonseed, whole	40.480	6.441	14.719	2.053	191.00	Chicago
Cottonseed meal, 41% CP	23.660	17.843	23.653	1.585	142.00	Chicago
Distillers dried grains	40.480	12.650	10.350	1.873	85.00	Chicago
Feather meal	36.387	56.036	24.015	1.488	170.00	Minneapolis
Fish meal, Menhaden	1.840	36.818	24.546	1.535	450.00	Chicago
Hominy feed	49.500	5.435	4.915	1.810	72.00	Chicago
Meat and bone meal	3.720	24.653	25.660	1.506	170.00	Chicago
Poultry byproduct meal	5.640	34.122	27.918	1.661	200.00	Kansas City
Soybean meal, solvent-extracted, 48% CP	7.200	17.357	32.234	1.810	136.00	Chicago
Tallow	0.000	0.000	0.000	5.840	200.00	Chicago
Wheat middlings	32.930	3.742	12.634	1.616	48.00	Chicago
Soybean hulls	60.970	4.625	6.386	1.608	60.00	Chicago
Bakery waste, dehydrated	16.560	2.412	7.432	1.896	70.00	Minneapolis
Cottonseed hulls	81.900	1.119	2.612	0.894	67.50	Chicago
Wheat grain, ground	10.680	3.275	10.965	1.812	95.00	Kansas City

[1]Composition data taken from NRC (12, 13) and Midwest market prices taken from Feedstuffs (6) are reported on an as-is basis.

Midwest – June 1999 – Initial Estimates

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Main results				
R-square	Adj. R-square	Root MSE	Dep Mean	C.V.
0.876	0.853	63.406	136.622	46.410

Analysis of variance					
Source	DF	Sum of square	Mean square	F Value	Prob>F
Model	4	597660.881	149415.220	37.165	0.0000
Error	21	84426.642	4020.316		
Total	25	682087.524			

Inference of coefficients					
Nutrient name	Estimate	Standard error	t-statistic	Prob.> |T|	Var. inflation
NDF	-0.027937	0.025907	-1.078	0.2931	1.7861
RUP	0.198323	0.044633	4.443	0.0002	2.6895
RDP	0.010147	0.073512	0.138	0.8915	4.1449
NEl	0.053797	0.013701	3.926	0.0008	3.1499

Correlation Matrix			
1	2	3	4
1.000	0.069	-0.043	-0.498
	1.000	-0.655	-0.007
		1.000	-0.494
			1.000

Calibration set							
Name	Actual [/T]	Predicted [/T]	Predicted-Actu	Leverage	Student res.	75.0% CI	75.0% CI
Barley, grain	87.498	91.086	3.588	0.046	-0.058	74.936	107.236
Blood meal	345.000	333.031	-11.969	* 0.552	0.282	277.306	388.756
Brewers grains, dehy.	72.000	106.249	34.249	0.083	-0.564	84.663	127.836
Corn gluten meal, 60%	210.000	230.031	20.031	0.103	-0.334	205.970	254.093
Corn gluten feed	55.000	87.771	32.771	0.095	-0.543	64.643	110.900
Corn grain, ground	78.570	99.969	21.399	0.071	-0.350	79.970	119.968
Soybean meal, solv, 44%	125.000	144.280	19.280	0.166	-0.333	113.716	174.844
Beet pulp, dehydrated	125.000	72.934	-52.066	0.113	0.872	47.703	98.164
Molasses, sugarcane	65.000	61.057	-3.943	0.045	0.064	45.230	76.885
Canola meal, solv-extd	96.000	108.119	12.119	0.134	-0.205	80.611	135.627
Cottonseed, Whole	191.000	106.132	-84.868	0.081	1.396	84.759	127.505
Cottonseed meal, 41%	142.000	139.719	-2.281	0.086	0.038	117.772	161.666
Distillers dried grains	85.000	121.073	36.073	0.070	-0.590	101.245	140.900
Feather meal	170.000	279.433	109.433	0.298	-2.060	238.466	320.400
Fish meal - Menhaden	450.000	224.906	-225.094	0.121	3.787	198.788	251.024
Hominy feed	72.000	83.248	11.248	0.114	-0.188	57.928	108.568
Meat and bone meal	170.000	174.420	4.420	0.108	-0.074	149.774	199.066
Poultry byproduct meal	200.000	218.896	18.896	0.126	-0.319	192.313	245.479
Soybean meal, solv, 48%	136.000	159.714	23.714	0.223	-0.424	124.320	195.108
Tallow	200.000	203.722	3.722	** 0.670	-0.102	142.338	265.106
Wheat Middlings	48.000	77.863	29.863	0.059	-0.486	59.565	96.161
Soybean Hulls	60.000	64.025	4.025	0.156	-0.069	34.358	93.692
Bakery waste, dehy	69.998	94.336	24.338	0.065	-0.397	75.257	113.416
Cotton, hulls	67.500	2.843	-64.657	0.352	1.267	-41.657	47.342
Wheat grain, ground	94.991	97.683	2.691	0.064	-0.044	78.774	116.591

* Feedstuffs with a high leverage on estimates.

** Feedstuffs with a very high leverage on estimates

SESAME: Regression results

Figure 4. Initial table of results as reported by SESAME (19) for the estimation of unit costs of NDF, RDP, RUP, and NE_L using market prices for 25 Midwest feedstuffs (6). This figure is an integral copy of the output from SESAME. Abbreviations are as follows: Adj R-square = adjusted R^2, Root MSE = square root of the mean squared error, Dep Mean = mean value of the dependent variable, C.V. = coefficient of variation, DF = degrees of freedom, Prob = probability, Var. inflation = variance inflation factor, Actual [/T] = observed price per metric ton, Predicted [/T] = predicted price per metric ton, Predicted-Actu = predicted minus observed price per metric ton, Leverage = leverage values, Student res. = studentized residuals, and 75.0% CI = lower and higher 75% confidence limits on predicted price.

Midwest – June 1999 – Final Estimates

18.12.1999 14:19:06

Main results				
R-square	Adj. R-square	Root MSE	Dep Mean	C.V.
0.965	0.956	23.722	102.978	23.036

Analysis of variance					
Source	DF	Sum of square	Mean square	F Value	Prob>F
Model	4	246177.477	61544.369	109.363	0.0000
Error	16	9004.046	562.753		
Total	20	255181.524			

Inference of coefficients					
Nutrient name	Estimate	Standard error	t-statistic	Prob.> |T|	Var. inflation
NDF	-0.002430	0.012544	-0.194	0.8488	2.6597
RUP	0.194114	0.040025	4.850	0.0002	5.2668
RDP	0.007716	0.043497	0.177	0.8614	8.2045
NEl	0.038849	0.009392	4.136	0.0008	6.9475

Correlation Matrix			
1	2	3	4
1.000	0.026	0.268	-0.719
	1.000	-0.713	-0.019
		1.000	-0.565
			1.000

Calibration set							
Name	Actual [/T]	Predicted [/T]	Predicted-Actu	Leverage	Student res.	75.0% CI	75.0% CI
Barley, grain	87.498	75.771	-11.727	0.122	0.528	65.874	85.669
Brewers grains, dehy.	72.000	107.478	35.478	0.174	-1.646	95.657	119.299
Corn gluten meal, 60%	210.000	207.103	-2.897	0.407	0.159	189.038	225.167
Corn gluten feed	55.000	83.753	28.753	0.151	-1.316	72.746	94.761
Corn grain, ground	78.570	79.989	1.419	0.233	-0.068	66.310	93.668
Soybean meal, solv, 44%	125.000	124.943	-0.057	0.212	0.003	111.895	137.992
Beet pulp, dehydrated	125.000	75.291	-49.709	0.138	2.257	64.777	85.805
Molasses, sugarcane	65.000	44.126	-20.874	0.142	0.950	33.464	54.787
Canola meal, solv-extd	96.000	99.147	3.147	0.212	-0.149	86.098	112.196
Cottonseed meal, 41%	142.000	127.640	-14.360	0.105	0.640	118.469	136.811
Distillers dried grains	85.000	114.754	29.754	0.114	-1.333	105.193	124.316
Hominy feed	72.000	83.254	11.254	0.145	-0.513	72.473	94.034
Meat and bone meal	170.000	152.571	-17.429	0.154	0.799	141.469	163.673
Poultry byproduct meal	200.000	195.026	-4.974	0.321	0.254	178.991	211.062
Soybean meal, solv, 48%	136.000	135.809	-0.191	0.278	0.009	120.883	150.735
Wheat Middlings	48.000	71.821	23.821	0.097	-1.057	63.010	80.632
Soybean Hulls	60.000	72.629	12.629	0.171	-0.585	60.927	84.332
Bakery waste, dehy	69.998	76.514	6.515	0.187	-0.305	64.281	88.747
Cotton, hulls	67.500	32.277	-35.223	0.470	2.039	12.868	51.686
Wheat grain, ground	94.991	77.750	-17.242	0.168	0.797	66.151	89.348

* Feedstuffs with a high leverage on estimates.

** Feedstuffs with a very high leverage on estimates

SESAME: Regression results

Figure 5. Final table of results as reported by SESAME (19) or the estimation of unit costs of NDF, RDP, RUP and NE_L using market prices for 20 Midwest feedstuffs (6). This figure is an integral copy of the output from SESAME. Abbreviations are as follows: Adj R-square = adjusted R^2, Root MSE = square root of the mean squared error, Dep Mean = mean value of the dependent variable, C.V. = coefficient of variation, DF = degrees of freedom, Prob = probability, Var. inflation = variance inflation factor, Actual [/T] = observed price per metric ton, Predicted [/T] = predicted price per metric ton, Predicted-Actu = predicted minus observed price per metric ton, Leverage = leverage values, Student res. = studentized residuals, and 75.0% CI = lower and higher 75% confidence limits on predicted price.

reduced collinearities. Alternatively, the nutrient associated with the largest VIF can be removed from the model. The remaining parameters must then be reestimated using the reduced **X** matrix.

EXAMPLE

Table 2 reports the composition of 25 feedstuffs (12, 13) and Midwest market prices (6) in June 1999. The nutrients selected for the estimation were: NE_L, RDP, RUP, and NDF. Results of the first estimation, as reported by a specialized Windows application, SESAME (19), are shown in Figure 4. The intercept estimate was not significant (B_0 = 40.67, SE = 71.54, P = 0.58) and was removed from the model in a preliminary step. All VIF were less than 5.0, indicating a well-conditioned **X′X** matrix. Of the 25 feedstuffs selected, tallow showed an excessive leverage value and was removed from the calibration set. Likewise, fishmeal had a large studentized residual and was also removed from the calibration set. In a subsequent iteration, the studentized residuals or leverage values of blood meal, feather meal, and whole cottonseed became excessive, resulting in their deletion from the calibration set.

Results from the final iteration are shown in Figure 5. The model explains a large proportion of the cost variance (uncorrected for the mean) with an $R^2 = 0.965$. Estimates of nutrient costs were as follows: NDF, –$0.002 (±0.013) per kg; RUP, $0.194 (±0.040) per kg; RDP, $0.008 (±0.043) per kg; and NE_L, $0.039 (±0.009) per Mcal. Both NDF and RDP costs were not different from zero and could be removed from the model. The strong negative correlations between RUP and RDP costs and between NDF and NE_L costs explain the magnitude of VIF, although all VIF were below a value of 10. SESAME calculates various statistics for all feedstuffs in the calibration set. These include: breakeven prices (labeled as "Predicted"), residual values (labeled as "Predicted-Actual"), leverage values (labeled as "Leverage"), studentized residuals (labeled "Student res."),

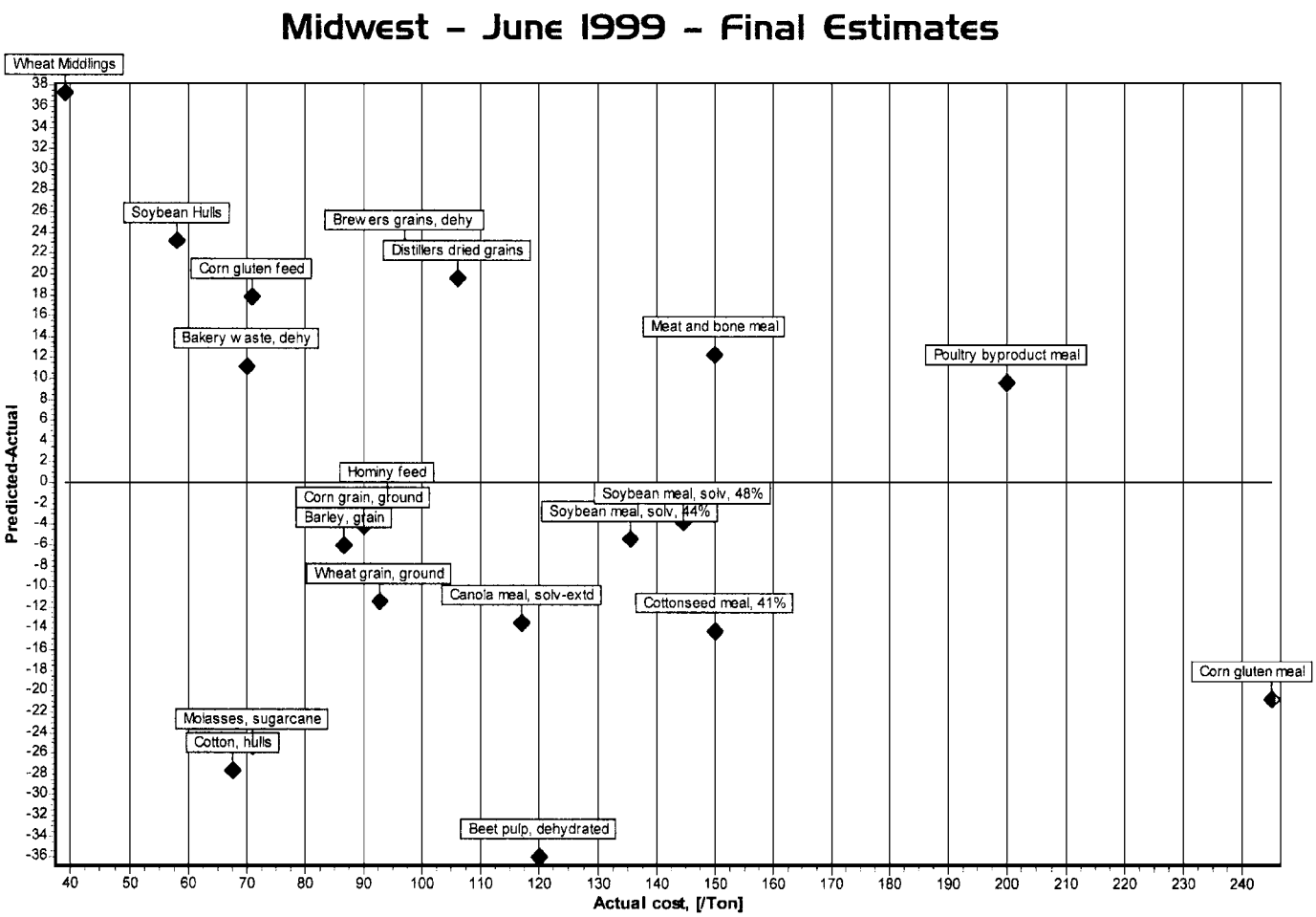

Figure 6. Residual plot from the estimation of unit costs of NDF, RDP, RUP, and NE_L for 20 Midwest feedstuffs (6).

and lower and upper 75% confidence interval of the breakeven prices (labeled "75% CI"). Deviations from estimated breakeven prices are also presented in a simple X-Y graphic of residuals versus actual costs of feedstuffs (Figure 6). Feedstuffs appearing below the zero residual line are relatively overpriced, whereas those appearing above the line are relatively underpriced.

CONCLUSIONS

The ML method uses the price of all feedstuffs traded in a given market to estimate the implicit costs of nutrients. Because it is a statistically based method, ML provides measures of dispersion of estimated nutrient costs and breakeven prices. Also, because the ML method does not use referee feeds, each feedstuffs used in the estimation can potentially have a breakeven price above or below its market price. The method has been programmed into a Windows application available from the authors. The long-run properties of ML estimates compared to PM remain to be determined.

REFERENCES

1 Beneke, R. R., and R. Winterboer. 1973. Linear Programming Applications to Agriculture. The Iowa State University Press, Ames, IA.

2 Cook, R. D., D. M. Hawkins, and S. Weisberg. 1992. Exact iterative computation of the robust multivariate minimum volume ellipsoid estimator. Stat. Prob. Lett. 16:213–218.

3 Dantzig, G. B. 1960. Inductive proof of the simplex method. IBM J. Res. Dev. 4:505–506.

4 Dennis, J. E., and R. B. Schnabel. 1996. Numerical Methods for Unconstrained Optimization and Nonlinear Equations. S.I.A.M., Philadelphia, PA.

5 Draper, N. R., and H. Smith. 1981. Applied Regression Analysis. John Wiley and Sons, New York, NY.

6 Feedstuffs. 1999. Ingredient Market. 71(3):50.

7 Heilbroner, L. R., and L. C. Thurow. 1978. The Economic Problem. 5th ed. Prentice-Hall, Englewood Cliffs, NJ.

8 Hoaglin, A. E., and R. E. Welsch, 1978. The hat matrix in regression and ANOVA. Am. Statist. 32:17–22.

9 Hoerl, A. E., and R. W. Kennard. 1970. Ridge regression: biased estimation for nonorthogonal problems. Technometrics 12:55–67.

10 Mason, R. L., and R. F. Gunst. 1985. Outlier-induced collinearities. Technometrics 27:401–407.

11 Morrison, F. B. 1956. Feeds and Feeding. 22nd ed. The Morrison Publishing Company, Ithaca, NY.

12 National Research Council. 1996. Nutrient requirements of beef cattle. 7th rev. ed. Natl. Acad. Sci., Washington, DC.

13 National Research Council. 1989. Nutrient requirements of dairy cattle. 6th rev. ed. Natl. Acad. Sci., Washington, DC.

14 Petersen, J. 1932. A formula for evaluating feeds on the basis of digestible nutrients. J. Dairy Sci. 15:293–297.

15 Rocke, D. M., and D. L. Woodruff. 1996. Identification of outliers in multivariate data. J. Am. Stat. Assoc. 91:1047–1061.

16 Rousseeuw, P. J., and A. M. Leroy. 1987. Robust Regression and Outlier Detection. John Wiley and Sons, New York, NY.

17 Schoeff, R. A., and P. Lobo. 1998. USA Feed Market. Feed Manage. 49(10):7–16.

18 Snedecor, G. W., and W. G. Cochran. 1980. Statistical Methods. The Iowa State University Press, Ames, IA.

19 St-Pierre, N. R. 1999. A User's Guide to Sesame. Dept. Animal Sciences, The Ohio State University, Columbus, OH.

20 St-Pierre, N. R., and W. R. Harvey. 1986. Uncertainty in composition of ingredients and optimal rate of success for a maximum profit total mixed ration. J. Dairy Sci. 69:3074–3086.

Sesame 3.5 Basic Tutorial

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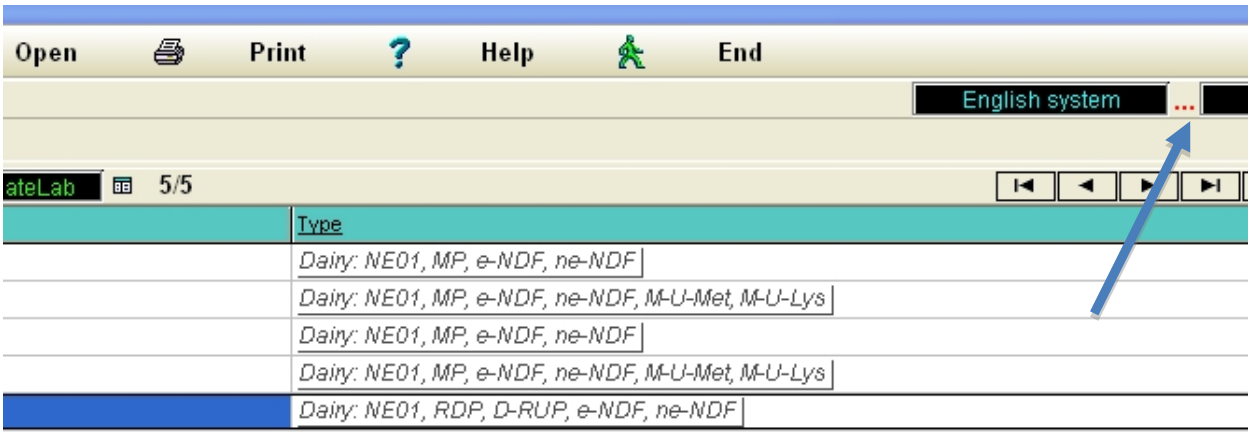
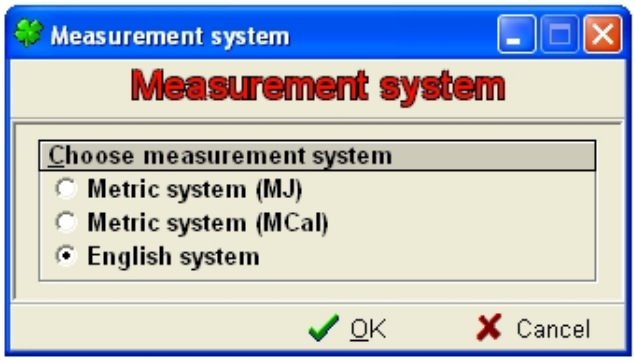
Problem 1

Feed prices have been fluctuating a lot in the last few years. You want to evaluate feedstuffs for the Midwest region based on their NE_L, metabolizable protein (**MP**), effective NDF (**eNDF**), and non-effective NDF (**neNDF**) content. You assume that the feeds have the same nutritional composition as those in the NRC 2001.

Click [*Solver*], then click on *Generic Problems*, and finally on *Midwest - Generic [3]*. In the *Solver* menu (top), click on [*Database*], choose [*Copy Problem*], and give it a new name, e.g. *workshop 1*. Click on *User Problems* in the top left window and you should see the new problem. At this point we will not add any new feeds to the problem. Make sure that you are using the English unit system. This is done by clicking on the […] button, as shown below.





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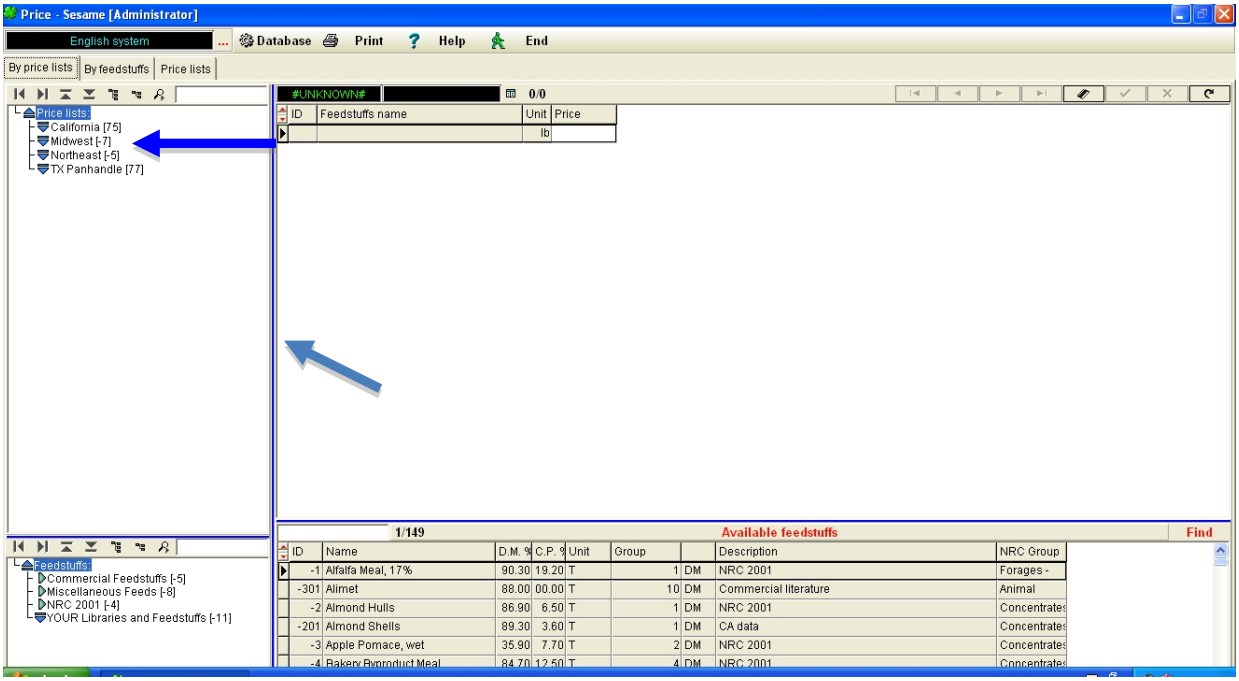
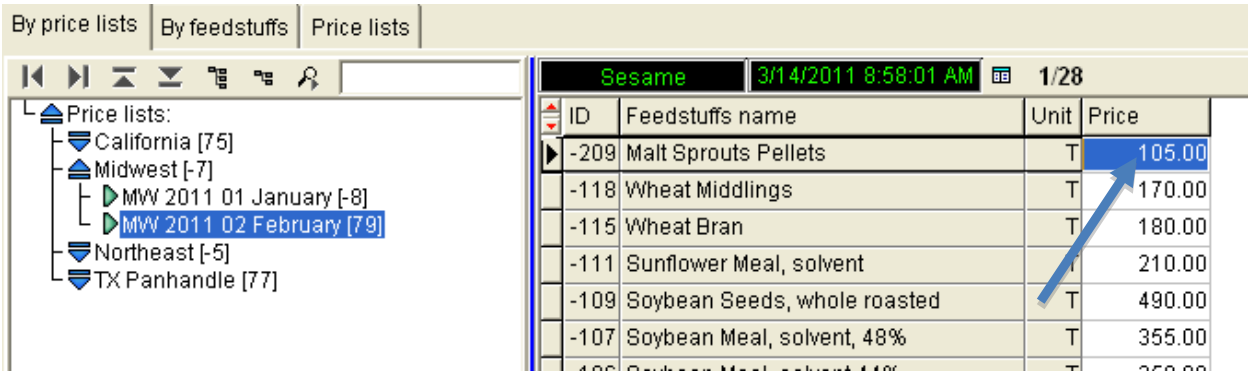
There are two ways to edit feed prices: 1) they can be edited directly in the *Solver* screen, or 2) they can be edited from *Price lists* in the main menu. If you are going to be running the same problem repeatedly over time, it is simpler to keep the same problem, and just update the price list by copying it. Price lists can be accessed directly from the main menu (planet earth), or via a short cut in *Solver*. We will use the shortcut here. Click the [*Price lists*] button in the *Solver* menu. This is a screen shot of the *Price lists* display when it opens:



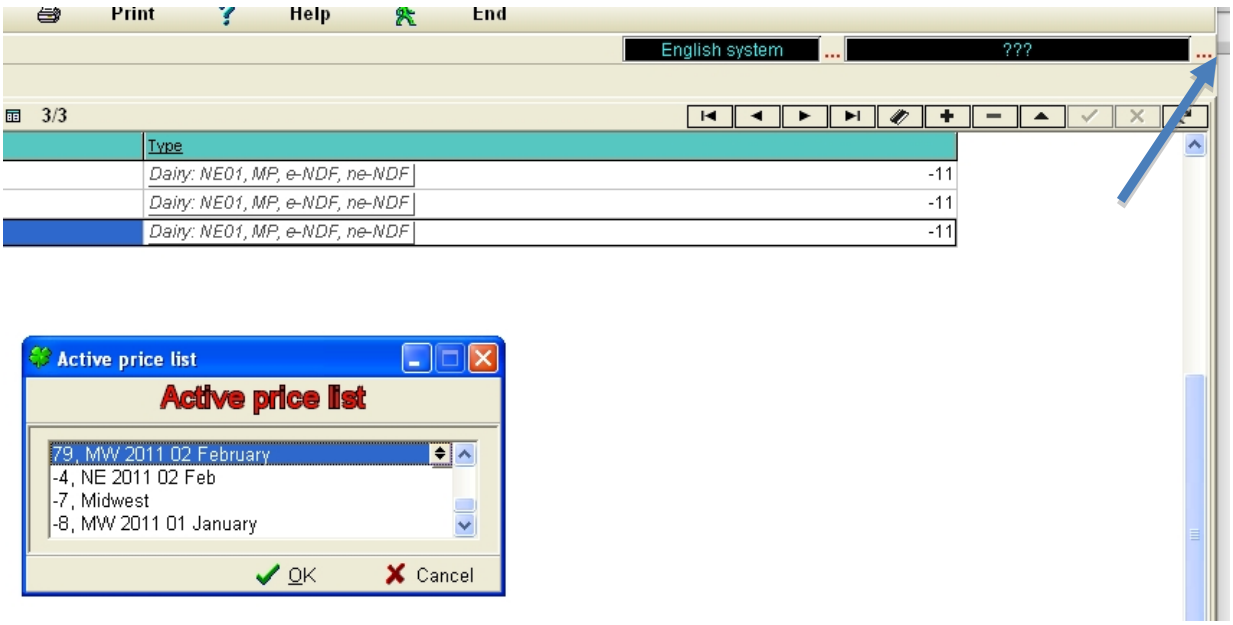
Note that the blue dividing lines separating the display into 4 windows can be moved to fit your screen; this is true throughout Sesame. We have already prepared a few price lists for you and arranged them in four libraries. Double-click the [Midwest] price list library (where the dark blue arrow is in the figure above). The display expands to show all price lists belonging to this library. In this instance, there is only one price lists: *MW 2011 01 January [-8]*. We will use this list as a starting point. First select the *MW 2011 01 January [-8]* price list by clicking on it in the top left window. Then, click on [*Database]* -> [*Copy Price list Library*]. Sesame opens a small window and suggests a default price lists name. The default would work, but we will select a name that will make the listing of our price lists much tidier. In the *Price List Name* window, overwrite the default name with: *MW 2011 02 February*. This new price lists is automatically added to the tree window (top left) in the Midwest library. Price lists in this tree window are sorted alphabetically. Thus, *MW 2011 02 February* comes after *MW 2011 01 January*. Using a code of Region--Year--Month--Month-alpha to name our new price list ensures a logical (time series) sorting.

This is where you could update all your feedstuffs prices. In this example, we will update only the malt sprouts price to keep this tutorial short. Change its price from $105 to $135/ton by clicking and entering the new price in the price cell.

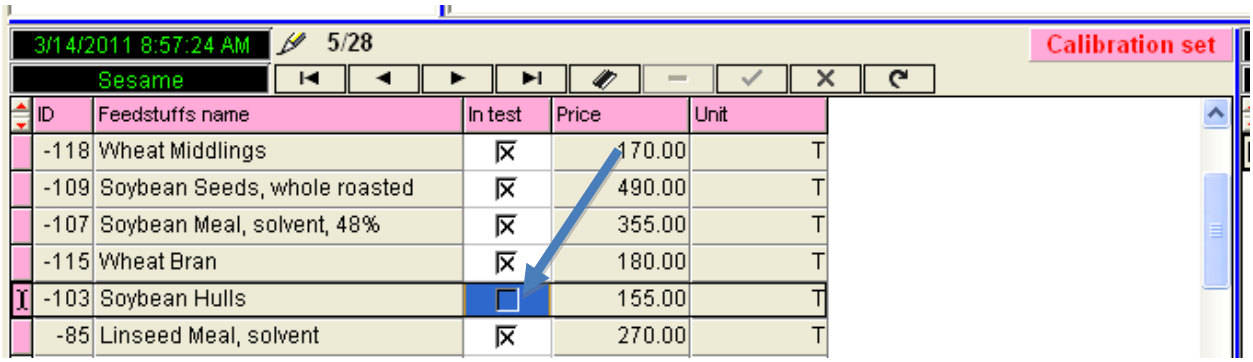
Note: Sesame operates on a database platform. An entry is not saved in the database until the cursor has moved to a new record. In this example, simply pressing [return] moves the cursor to the next record, but this would not necessarily occur everywhere in Sesame. It is always a good idea to click the button in the menu bar (top right) to refresh the database.



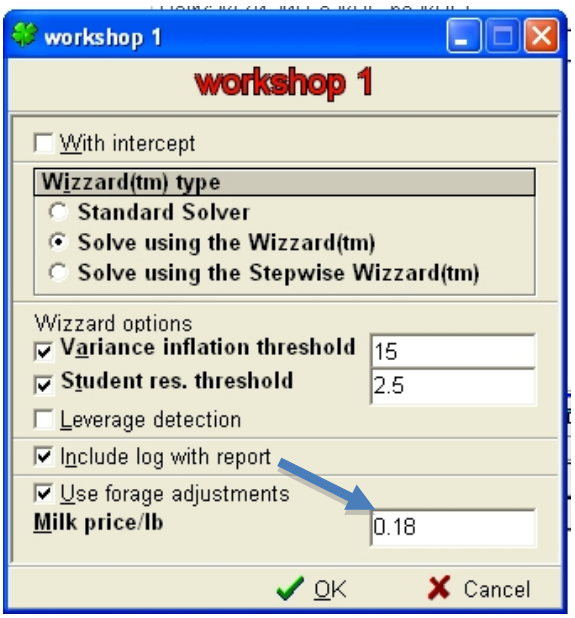
Now you are ready to solve the problem. Return to *Solver* by clicking [*End*] in the *Price* module menu. This brings you back directly to the *Solver* module. Select *workshop 1* under the *Users Problems* in the upper left hand window. You will see in the upper right hand window that this problem is already set to solve on NEl, MP, eNDF, and neNDF. The feedstuffs are in the lower left-hand window, but the prices haven not been updated yet. Select the correct price list by clicking on "**…**" in the upper right hand corner and choosing *MW 2011 02 February* on the pull-down list.



For some reasons, soybean hulls are not available; we would like to take this feed out of the problem. It can either be (1) permanently deleted from the problem by clicking the '-' button in the task bar, or (2) temporarily eliminated from the test by un-checking the "in-test" box, or (3) by moving it from the *Calibration* to the *Appraisal* set (just click-drag-and drop from the *Calibration* to the *Appraisal* window). In this example, we picked option (2).



Choose *Solve Problem* from the top menu. This opens the Wizzard™ window. The default solver options are to Solve using the Wizzard with a Variance inflation threshold of 15 and a Student residual threshold of 2.5. Generally, you don't want to change these options, so we will use the defaults and change only the milk price from 0.14 to 0.18/lb. The milk price is important for calculating the price adjustments of forages. Press [OK] to solve the problem.



The first section of the report shows the Price Prediction Reliability (**PPR**) and the estimates of nutrient costs. The PPR is a measure of the error in the model. As satisfactory solution should have a PPR less than 20% of the average feed price in the problem. In this problem, the average price was $260.50. The PPR of 39.54 is only 15.2% of the average price, so the solution should be very good.

Note: How do we know what is the average price? There are two buttons at the bottom of the solution screen. Select '*Full report*' and the number appearing under '*Dep Mean*' in the *Main results* table is the average feed price.

workshop 1

Price Prediction Reliability	39.944

Estimate of Nutrient Unit Costs

Nutrient name	Estimate	
NEI - 3X (2001)	0.086661	**
Metabolizable Protein (MP, g/	0.407165	**
e-NDF	0.066242	*
ne-NDF	-0.012095	

- A blank means that the nutrient unit cost is likely equal to zero
- ~ means that the nutrient unit cost may be close to zero
- * means that the nutrient unit cost is unlikely to be equal to zero
- ** means that the nutrient unit cost is most likely not equal to zero

The numbers in the '*Estimate*' column are the estimated costs per unit for each nutrient selected in the problem. In this problem, for example, net energy costs 8.67¢/Mcal while MP costs 40.7¢/lb.

Calibration set

Name	Actual [/T]	Predicted [/T]	Lower limit	Upper limit	Corrected	75.0% CI	75.0% CI
Bakery Byproduct Meal	270.000	212.157	192.820	231.494	-	-	-
Beet Sugar Pulp, dried	230.000	195.125	182.217	208.033	-	-	-
Blood Meal, ring dried	660.000	643.041	608.366	677.717	-	-	-
Canola Meal, mech. extracte	257.000	241.634	231.148	252.121	-	-	-
Corn Grain, ground, dry	240.000	214.658	196.745	232.571	-	-	-
Corn Silage, 35% DM	45.000	74.715	67.145	82.284	74.715	67.145	82.284
Cotton Seed Hulls	155.000	106.243	73.406	139.080	-	-	-
Cotton Seed Meal, 41% CP	295.000	313.981	302.138	325.825	-	-	-
Cotton Seed, Whole w lint	275.000	272.208	237.185	307.230	-	-	-
Distillers Dried Grains w Sol	195.000	266.964	250.214	283.714	-	-	-
Gluten Feed, dry	159.000	217.448	206.147	228.750	-	-	-
Gluten Meal, dry	558.000	524.202	501.064	547.340	-	-	-
Hominy	183.000	198.736	184.072	213.399	-	-	-
Leg Hay, all samples	160.000	182.305	159.111	205.499	203.154	179.960	226.348
Linseed Meal, solvent	270.000	258.908	245.886	271.931	-	-	-
Malt Sprouts Pellets	135.000	182.544	166.107	198.982	-	-	-
Meat Meal, rendered	340.000	375.999	357.818	394.179	-	-	-
Molasses, Sugarcane	180.000	143.595	127.452	159.739	-	-	-
Soybean Meal, expellers	380.000	431.979	416.320	447.638	-	-	-
Soybean Meal, solvent 44%	350.000	309.233	296.211	322.256	-	-	-
Soybean Meal, solvent, 48%	355.000	355.300	341.142	369.458	-	-	-
Sunflower Meal, solvent	210.000	161.642	144.861	178.423	-	-	-
Wheat Bran	180.000	163.962	144.597	183.327	-	-	-
Wheat Middlings	170.000	180.387	163.570	197.205	-	-	-

The *Calibration set* table reports the results for the feeds that were in the calibration set (i.e., those feeds for which we had both a composition and a market price). The numbers in the columns have the following meanings: *Actual[/T]* are the prices that we entered in the problem; *Predicted[/T]* are the estimated break-even prices; *Lower limit* are the 12.5% lower confidence intervals on the estimated prices; *Upper limit* are the 87.5% upper confidence intervals on the estimated prices; *Corrected* are only reported for forages – they include correction factors on the estimated values of forages based on the work of Weiss and St-Pierre; the last two columns labeled *75.0% CL* are the 75% confidence limits on the corrected prices. Feeds with an actual price/ton less than the *Lower limit* are deemed bargains; those with an actual price exceeding the *Upper limit* should be considered over-priced.

The last section of the report shows the results for the *Appraisal set* and also displays a *Calculation log*. When we submitted the problem there were no feeds in the *Appraisal set*. But while solving, Sesame determined that the actual prices of Fish Menhaden Meal, Soybean Seeds, whole roasted, and Tallow were so different from their calculated values that they should be deemed outliers. In such instances, Sesame automatically moves the outlier feeds to the *Appraisal set*, where they have no impact whatsoever on the solution. This movement of outlier feeds from the *Calibration* to the *Appraisal* set is done sequentially. The sequence is reported in the *Calculation log*.

Appraisal set

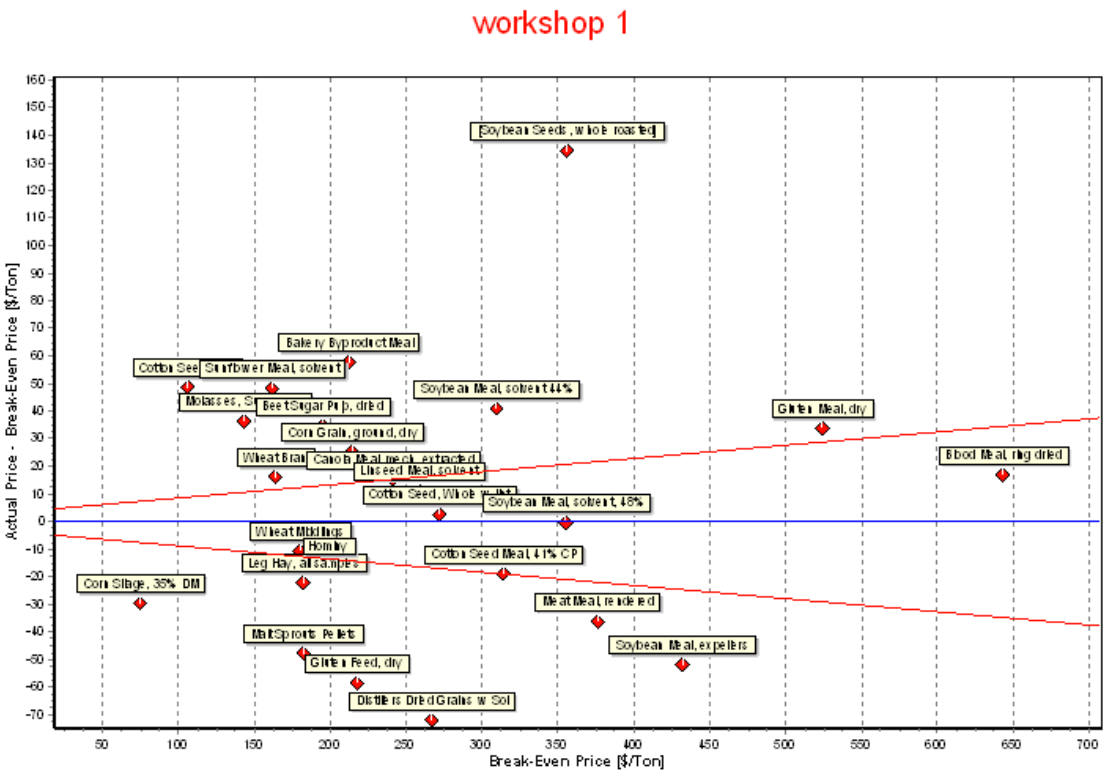
Name	Actual [/T]	Predicted [/T]	Corrected
Fish Menhaden Meal, mech.	1450.000	513.482	-
Soybean Seeds, whole roas	490.000	355.663	-
Tallow	980.000	447.193	-

Calculation log

Action	Reason
Feedstuffs "Fish Menhaden Meal, mech." removed	Outlier
Feedstuffs "Tallow" removed	Outlier
Feedstuffs "Soybean Seeds, whole roasted" removed	Outlier

Results can also be viewed in a graphic format. Click the [*Graph*] button in the *Report* menu. This opens a *Graph Style* window.

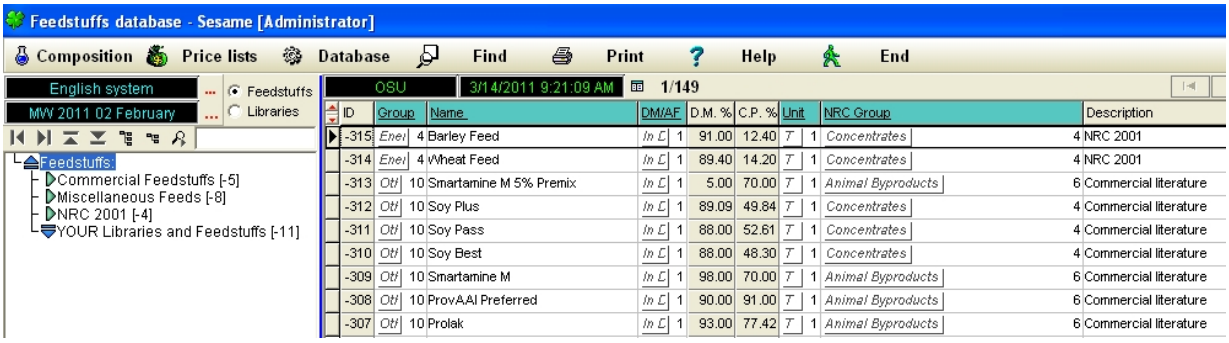
The second option is used to block the display of feeds that are substantially different than the average price. In this example, change this limit to 200%; then click OK. The following graph is displayed. Feeds above the top red line are over-priced. Feeds below the bottom red line are under-priced. Feeds between the two red lines are priced relatively close to their calculated break-even values. Beware of how you are interpreting these results. The results do not imply that a ration can be balanced using only bargain feeds, or that a ration can be balanced without using any of the over-priced feeds. But as one balances specific rations on a farm, a coherent strategy is to maximize the use of bargains feeds and to minimize the use of over-priced feeds while maintaining a balanced ration. Also, some feeds are used in rations for reasons other than their nutritional content in NE_L, MP, e-NDF, and ne-NDF. For example, molasses can be used to stick fine particles to large particles in the diet so as to avoid separation of the ingredients in the TMR. In our example, molasses is over-priced by $47/ton. The average cost of using molasses as a 'TMR glue' is thus ~ 2.35¢/lb of molasses.



Problem 2

Next you would like to look at the value of a defatted, high protein distillers dried grains. Begin by making a copy of Problem 1. In the *Solver* module, select *workshop 1* in the *User problems* library in the tree window (top-left window). In the *Solver* menu, click on *Database* > *Copy problem*) and name it *workshop 2*.

Next, we need to add the new DDGS feed to our feed library. Close the *Solver* module by clicking on [*End*]; this brings you to the *Main Menu* (planet earth). Select the *Feedstuffs* module (top left); this opens the *Feedstuffs* module. This module is used for creating, copying, or editing feedstuffs.



All feeds belong to the *Feedstuffs* library. In addition, they can be attached for convenience in either one of the 3 libraries that we created (*Commercial Feedstuffs, Miscellaneous Feeds,* or *NRC 2001*), or in your own libraries and feedstuffs (*Your Libraries and Feedstuffs).*

Click on the *NRC 2001* library (upper left window); then choose (click) on *-23 Distillers Dried Grains wSol* in the upper right window. Note that feeds are listed in ascending ID numbers.

> Note. You cannot edit the composition of feeds with a negative ID number (e.g., -315 for Barley Feed). These are 'protected' feeds. If you want to change the nutritional composition of a protected feed, you must first create a copy of the feed and edit the copy, which will have a positive ID number.

In the *Feedstuffs* module menu, select *Database* > *Copy Feedstuff*, and name the new feed *hi pro DDGS*. In the tree of libraries window (top left), click on *YOUR libraries and Feedstuffs*. Your newly created feed will appear in the *Feedstuffs* windows as ID=55, Name=hi pro DDGS.

> Note. The feed ID number for your copy might not be [55]. Don't worry if it is not. Sesame has its own way of numbering feeds to ensure that no two feeds share the same ID number.

Click on *hi pro DDGS* in the *Feedstuffs* (upper right) window, and then click on [*Composition*] on the top menu. This will open the *nutrient composition* window. In the upper left window, select *Basic Nutrients (Across Species).* Nutrients are grouped in sub-libraries for ease of entry. Double-clicking on any of the sub-libraries expands the display in both the *nutrient library* tree window (top left) and *nutrient* window (top right). For simplicity, we are going to assume that the process for producing this DDGS only affects crude protein and ether extract, not ash and NDF. In the *nutrient* (upper right) window change Crude protein from 29.7 to 40% and Ether extract from 10 to 5%. Notice that the NFE and NFC changed, since they are calculated values. Nutrients with an ☒ in the *Use formula* column are automatically calculated using equations stored in Sesame.

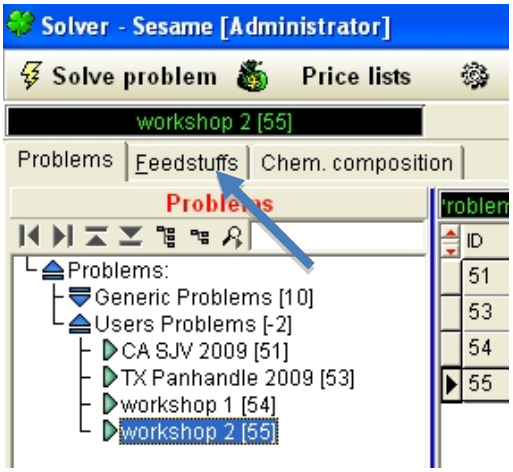
Note. What nutrients should be entered in Sesame? The answer depends on which nutrients will be selected in the problem. If the suggested nutrient group [-11] is used (NE_L, MP, e-NDF, and e-NDF) then the following nutrients must have a valid entry for the nutrients to be properly calculated (Sesame uses the NRC(2001) model for its calculation):

[-1]	DM	[-2]	CP	[-3]	Ether extracts
[-5]	Ash	[-8]	NDF	[-11]	Lignin
[-14]	NDICP	[-300]	RUP	[-301]	RUP digestibility
[-302]	NDF effectiveness				

Some feeds also require:
[-297] Processing adj. factor – used for some processed grains
[-298] True protein digestibility – required for animal proteins
[-299] True fatty acids digestibility – required by high fat ingredients

Do not forget to click on the refresh data button (upper right task bar) before leaving this screen or else your composition changes will not be saved. Click [*End*] to leave the *Composition* screen, followed by another [*End*] to leave the *Feedstuffs* module. You should be back to the main menu (planet earth).

Click *Solver* to access the *Solver* module. Click on *User Problems* and select *workshop 2* in the upper left window. The problem should have the same nutrient basis and feedstuffs as Problem 1. Click on the *Feedstuffs tab*.



This opens your feed libraries. Click on *Your Libraries and Feedstuffs* in the *feed library* window (top left). The ingredient *hi pro DDGS* now shows up in the *feedstuffs* (upper right) window. Click-drag-and-drop *hi pro DDGS* into the *Appraisal set* window (bottom right). We don't know what the price of this feed really is, which is why we put it in the *Appraisal set*. Had we known both its composition and price, we would have put it in the *Calibration set*. Although we do not know what its price really is, we think that it should be about $10/ton over regular DDGS. So we right click on the price cell to change its price to $205/Ton.

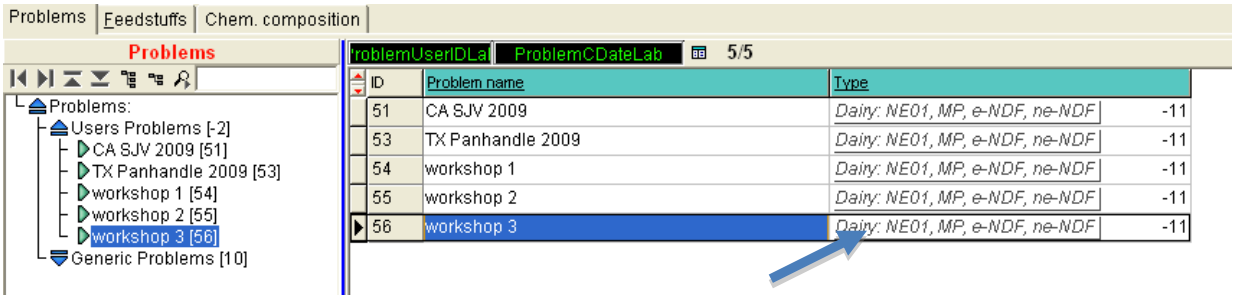
> Note. We are entering a new price only to illustrate how prices can be changed 'on the fly' in the *Solver*. The price of any feed in the *Appraisal set* has no impact whatsoever on the solution and, hence, its estimated break-even price.

Click [*Solve Problem*] from the top menu and go through the same steps as in problem 1. In the results, note that the average price of the feedstuffs (Dep mean) and the nutrient values are unchanged from Problem 1 since we did not change any feeds in the *Calibration* set. The predicted value of the defatted, high protein DDGS is $289.74/Ton.

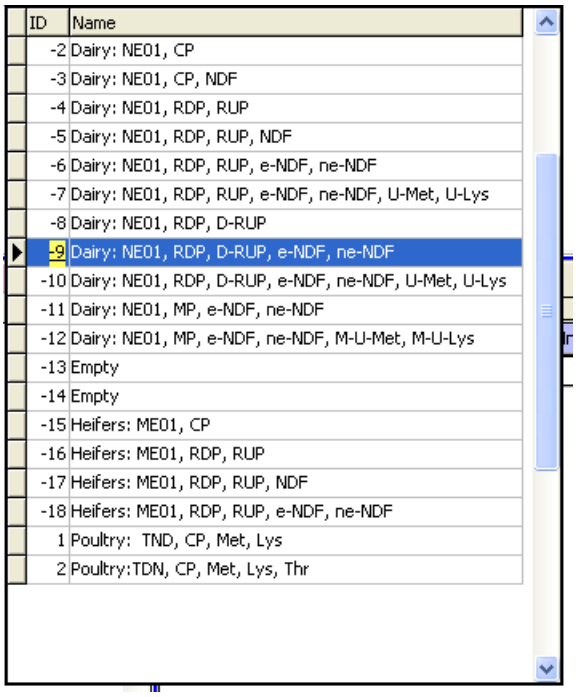
Problem 3

Instead of evaluating feeds based on MP, you want to evaluate them based on rumen degradable protein (**RDP**) and digestible rumen undegradable protein (**dRUP**). Begin by copying *workshop 2* as you did in Problem 2, naming the copied problem *workshop 3*.

Make sure that *workshop 3* is selected (i.e., click on *workshop 3*). Then, in the upper right window, double-click on the *Type* cell for workshop 3.



A new window will open with several pre-programmed choices.



Choose *-9 Dairy: NEl, RDP, D-RUP, eNDF, neNDF*. Then click the refresh button in the top right task bar to make sure that your changes have been saved.

> Note. It is possible to program your own nutrient choices, but this is a topic for an advanced Sesame user. You must learn to walk before you run…

Click on *Solve problem*, and keep all the same defaults for solving as what were used in problems 1 and 2.

Price Prediction Reliability	48.630

Estimate of Nutrient Unit Costs		
Nutrient name	Estimate	
NEI - 3X (2001)	0.129850	**
RDP	-0.017684	
Digestible RUP	0.374633	**
e-NDF	0.057232	~
ne-NDF	-0.016103	

- A blank means that the nutrient unit cost is likely equal to zero
- ~ means that the nutrient unit cost may be close to zero
- * means that the nutrient unit cost is unlikely to be equal to zero
- ** means that the nutrient unit cost is most likely not equal to zero

Notice that the Price Prediction Reliability is greater than what we had in problems 1 and 2 when we solved for MP as opposed to RDP and dRUP. In this instance, solving for MP is better than solving for RDP and dRUP. It is not always like that. Some people prefer solving for MP, arguing that the protein requirements of a cow are best expressed in terms of MP, while others prefer RDP and dRUP, arguing that these 2 nutrients can be measured, while MP is entirely calculated – even in research. As long as you maintain the same group of nutrients, feed comparison across time should be valid. Also, generally there are little meaningful differences between the calculated break-even prices using either set of nutrients.

Summary

In this tutorial, you learned how to:

1. Set up a problem with a Calibration set of feedstuffs and an appropriate price list.
2. Copy and Edit price lists.
3. Copy and Edit feedstuffs.
4. Change the nutrient values for the basis of the analysis.
5. Interpret Sesame 3.5 reports.

Keep exploring Sesame on your own. Periodically, **you should back-up your data files** in case something happens to your system, or if things become so messed up that you want to start back where things were working. To backup your data, select [*Backup*] on the main menu (planet earth), then select *backup*. Sesame will automatically create a .zip file containing all your data and will name it based on the date and time of your backup. Don't forget to make a copy of the backup files on something else than your hard drive because you don't want to loose all your work if your hard drive ever fails (and they all do… eventually).

Economics of Making Nutritional Decisions with Volatile Feed Prices

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INTRODUCTION

Historically, attractive corn prices coupled with its wide availability have led to heavy corn utilization as the main source of energy in dairy feeding programs in the United States. In practical dairy nutrition, this has resulted in corn grain being used to maximize levels and amounts of fermentable carbohydrate that provides energy to both the rumen microbes and to the dairy cow. As a consequence, recent approaches in balancing dairy rations have focused on the optimal level of dietary starch to optimize rumen fermentation and lactation performance (Emanuele, 2005; Grant, 2005).

With the advent of higher corn prices and concurrent price increases in other grains and many by-products, questions have been raised regarding corn use in dairy diets, and alternative feeding and ration balancing strategies with better economic outcomes. The economic need forces the necessity to re-evaluate the role and value of carbohydrates in dairy rations, as well as ration formulation strategies to reduce feed costs. In this paper, we analyze the need for corn in dairy diets; challenge the dogma of a narrow optimal range for dietary starch and non-fiber carbohydrates (**NFC**) in dairy rations, compare current feed and production economics to those that prevailed a few years ago, and explain how maximum economic feed efficiency can be attained.

COWS DO NOT REQUIRE CORN

Some have mistakenly equated the wide use of corn in dairy rations to a requirement for corn by the animals. This is incorrect both nutritionally and economically. Table 1 summarizes the results of 2 experiments where barley was substituted for corn in lactating cow rations. None of the production, composition, intake, and feed efficiency parameters were significantly affected by the type of grain used. Although some small numerical differences would seem to favor corn, it is important to remember that these studies were designed to maximize production differences between diets. For example, in the experiment of Bilodeau et al. (1989) the substitution was entirely on an ingredient basis. That is, 43.7 % of the diet was either corn or barley without any other dietary adjustments to make the diets isocaloric. Thus, the barley diet had only 97 % of the caloric density of the corn diet.

LITTLE EVIDENCE FOR NARROW OPTIMUM LEVELS OF STARCH AND NFC

Many nutritional rules of thumbs were derived during times when corn was a very inexpensive feed ingredient. Some of these rules led to good working rations that were economically efficient when corn was cheap. For example, some have recommended constraining dairy diets to 25 to 30 % starch and 35 to 40 % NFC. The

Table 1. Effects of feeding corn or barley on milk production, composition, dry matter intake, and feed efficiency.

	DePeters and Taylor, 1985			Bilodeau et al., 1989		
	Corn	Barley	*P*	Corn	Barley	*P*
Yield (kg/d)						
Milk	28.0	27.4	n.s.	29.2	28.9	n.s.
Fat	0.835	0.761	n.s.	-	-	n.s.
Protein	0.893	0.884	n.s.	-	-	n.s.
Composition (%)						
Fat	3.01	2.81	n.s.	3.98	3.96	n.s.
Protein	3.21	3.23	n.s.	3.36	3.34	n.s.
Dry matter intake (kg/d)	18.5	18.3	n.s.			n.s.
Gross feed efficiency	1.51	1.50	-	1.33	1.31	-

experimental evidence to substantiate these recommendations is very thin. What has been lost is that rumen microbes do not have a requirement for starch per se; the energy requirement can be satisfied by fermentable carbohydrate derived from the hydrolysis of either NFC or neutral detergent fiber (**NDF**). Thus, significant amounts of by-product feeds can be used as replacements for corn without much effect on animal productivity.

By-product feeds can be used to replace both forages and grain in dairy cattle diets. Several research articles have been published on the ability of by-products such as corn gluten feed, beet pulp, and soy hulls to replace forage (Boddugari et al., 2001; Clark and Armentano, 1997; Ipharraguerre & Clark, 2003). Alternatively, and more importantly in this era of ethanol euphoria, by-products can also be used to replace grains (Beckman and Weiss, 2005; Boddugari et al., 2000; Ipharraguerre et al., 2002; Voelker & Allen, 2003a). By-products are generally much lower in starch than grains; but contain significant quantities of other NFC, including sugars, organic acids, fructans, glucans, and pectins. These sources of NFC are generally very degradable in the rumen and can provide energy to both the rumen microbes and to the cow. Several research studies have shown no decrease in rumen microbial flow to the small intestine, total tract NFC and NDF digestibilities, dry matter intake (**DMI**), and milk yield and milk components when by-product feeds are substituted for corn grain in dairy rations (Table 2). Starch contents of the diets ranged from 9.2 to 38.3 % DM, with corresponding NFC levels ranging from 27.2 to 50.7 % of DM and NDF levels inversely ranging from 49.4 to 24.3 % of DM. In all of these studies forage NDF levels were maintained within or above current NRC recommendations for forage NDF.

A quadratic response function of milk yield to dietary starch was fitted for each study reported in Table 2. The point of maximum production was estimated by equating the first derivative of this function to 0 and solving for starch. Maximum milk production was achieved at starch levels of 24.4, 33.3, 28.0, and 40.7 % of DM for each of the 4 studies. Clearly, this is a very wide range of optima. In addition, the response functions of milk to dietary starch are very flat, indicating that milk production does not respond very much to dietary starch over a range of 15 to 40 % of DM. The recommended 25 to 30 % range for dietary starch may have resulted in good working rations in the past, but the economic penalty for this myopic view is now, as we shall see, excessive.

Table 2. Summary of selected published research demonstrating that replacement of corn grain in lactating cow diets with other sources of fermentable carbohydrate does not impact milk production and can increase feed efficiency. Nutrients are reported on a DM basis. Abbreviations used: WCGF-wet corn gluten feed, SBM-soybean meal, SH-soy hulls, CSH-cottonseed hulls, and HMC-high moisture corn.

Reference	Diet as described in reference	Forage NDF	Total NDF	NFC	Starch	Milk yield (lb)	DMI (lb)
Boddugari et al., 2001	0	22.8	28.2	43.2	30.3	66.9	54.3
WCGF vs. corn + SBM	50	22.8	35.4	36.5	23.3	67.1	49.5
(% WCGF)	75	22.8	38.2	32.9	18.9	67.8	50.8
	100	22.8	41.6	29.4	15.1	64.9	48.0
Beckman & Weiss, 2005	35	18.2	24.7	48.3	33.3	71.1	44.7
SH+CSH vs. corn	29	18.2	28.6	43.7	30.1	69.7	46.2
(% corn)	23	18.2	32.2	40.4	25.4	69.5	47.7
Ipharraguerre et al., 2002	0	19.1	29.4	50.7	38.3	64.9	52.4
SH vs. corn	10	19.1	34.4	44.8	31.1	64.5	54.6
(% SH)	20	19.1	39.9	39.0	23.8	65.8	53.7
	30	19.1	44.8	33.1	16.5	64.5	50.4
	40	19.1	49.4	27.2	9.2	62.3	49.9
Voelker & Allen, 2003a	0	17.1	24.3	47.0	34.6	80.1	54.6
beet pulp vs. HMC	6	17.1	26.2	45.0	30.5	80.5	55.0
(% beet pulp)	12	17.1	28.0	43.0	26.5	79.0	55.2
	24	17.1	31.6	39.1	18.4	77.9	50.4

A similar method was used to estimate the optimum NFC level in each of the 4 studies listed in Table 2. Maximum milk production was achieved at NFC levels of 37.9, 33.3, 42.4, and 49.7 % of DM across the 4 studies. As for dietary starch, it appears that the optimal NFC level for milk production is poorly defined, and that milk production is not very responsive to NFC level in the diet. Again, a recommendation of 35 to 40 % NFC in dairy rations may have resulted in good, practical rations in the era of cheap feedstuffs, but constraining diets within this narrow range is likely no longer economically optimal.

WHY DAIRY COWS EXPRESS A SMALL RESPONSE TO STARCH

Significant attention has been given to rumen degradation rates of dietary carbohydrate fractions and synchronization of rumen degradable protein with carbohydrates. This has culminated in an emphasis on these rates in static nutrition models currently used in the field. Static models estimate nutrient availability at a single time point; they do not integrate over time as dynamic models do.

While the carbohydrates in NFC (sugars, organic acids, fructans, glucans, pectins, and starch) and in NDF (hemicellulose and cellulose) have different degradation rates from each other which can vary under different conditions; the degradation products of all of these supply a single rumen carbohydrate pool that is potentially fermentable by microbes, designated as the fermentable carbohydrate (**FC**) pool here. Because cows eat multiple meals/d and there are multiple carbohydrate fractions that contribute to the FC pool, this pool approaches steady-state kinetics under common dairy management practices (Figure 1A). Altering the proportion of starch vs. other carbohydrates does not change the flow into the FC pool (Diet A vs. Diet B). The FC pool is rapidly fermented by the rumen microbes, providing energy for their maintenance and growth, and VFA to the cow for energy and milk precursors. Consequently, varying dietary starch content does not change the energy availability to the microbes or the cow, given that total NFC is maintained and that the carbohydrate fractions are degradable. This conclusion is supported by experimental evidence that demonstrates no change in microbial protein flow to the small intestine when dietary starch content is varied (Voelker and Allen, 2003b; Ipharraguerre et al., 2005).

This lack of difference between diets with altered NFC components also holds true under non-steady-state conditions (Figure 1B), such as slug feeding during heat stress situations. The pool of each carbohydrate will vary over the course of the day as a function of meal size and meal intervals, but as long as the total NFC and NDF levels are maintained, there is no difference in replacing starch with other NFC. The model assumes that passage of the carbohydrate fractions out of the rumen is not affected by the meal feeding pattern. Note that during the 12-hr period when the cow is eating, the FC pool approaches steady-state (19 hr to 7 hr). Replacing the starch in corn grain with equal amounts of other NFC or digestible NDF in by-product feedstuffs does not change energy availability, despite differences in rumen degradation rates between carbohydrate fractions.

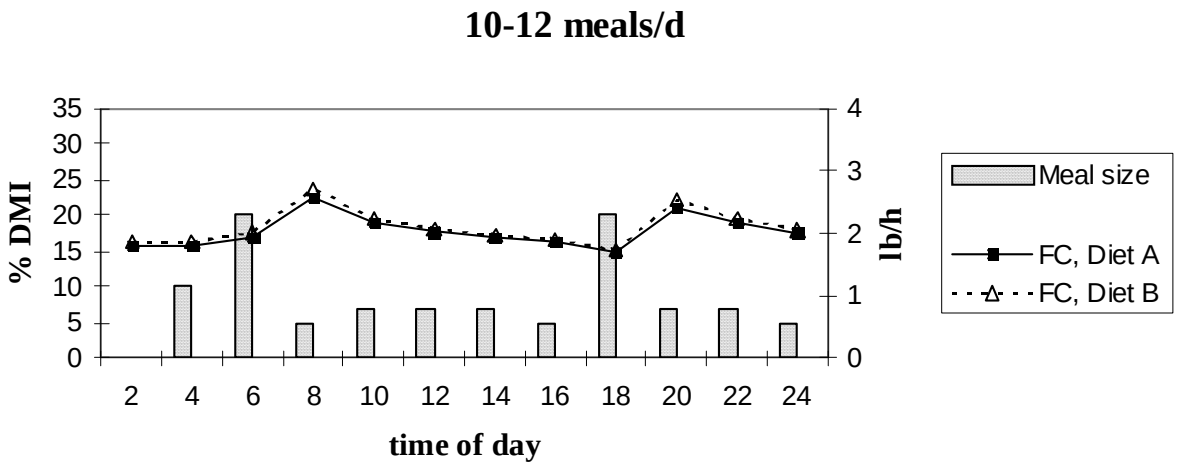
COMPARING AND EVALUATING FEEDSTUFFS

A simple approach to evaluating feedstuffs is to compare their cost/unit of energy and protein. Energy can be based on total digestible nutrients (**TDN**) or net energy for lactation (**NE_L**). Traditionally, prices/unit of energy and crude protein (**CP**) were based on the cost of corn and soybean meal, respectively. This approach, first proposed by Petersen (1932) and found in most applied nutrition books, such as that of Morrison (1956), assumes that these 2 feeds are perfectly priced. That is, the assumption here is that their selling prices are always equal to the economic value of their nutrients. This assumption doesn't make much economic sense as it would imply perfect and efficient markets. That this doesn't hold has been very evident with the current erratic grain and protein prices. The simple energy/protein approach also ignores other nutrients that are important in ruminant nutrition.

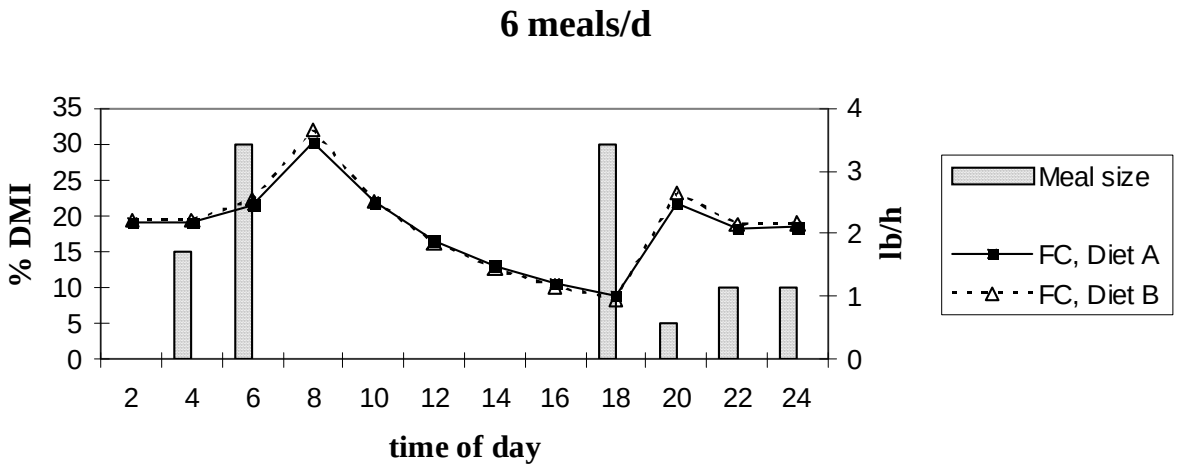
Additional approaches have been developed to include consideration of multiple nutrients. Increasing nutritional costs are captured in FEEDVAL (Howard & Shaver, 1997), which evaluates feedstuffs based on CP, TDN, calcium, and phosphorus

Figure 1. Energy availability to rumen microbes is not altered by varying starch content while maintaining NFC with other sources of fermentable carbohydrate (**FC**) under either semi-steady state (10-12 meals/d, panel A) or non-steady state (6 meals/d, panel B) feeding conditions. Diet A contains 30 % NDF, 40 % NFC, 25 % starch, and Diet B contains 30 % NDF, 40 % NFC, 15 % starch.

A



B



using 4 reference feedstuffs. Earlier versions have also allowed evaluations based on RUP. Because the reference feedstuff for energy is shelled corn, the value of energy predicted by this program will be high when corn prices are high. Fundamentally, the approach is nothing more than an expansion of the Petersen method for more than 2 nutrients and, thus, suffers the same limitations as the Petersen method.

A method that uses prices and nutritional composition from all feedstuffs traded in a given market has been proposed by St-Pierre and Glamocic (2000). The method uses a

multiple regression approach to set as many equations as there are feedstuffs. Estimates of unit costs for each important nutrient are obtained by least-squares. The resulting software, ***Sesame III*** is a Windows-based program and is available at www.sesamesoft.com. This program uses a multiple regression approach to estimate break-even prices of a set of feedstuffs based on their nutrient contents and market prices. Consequently, it can be used to determine the relative price of individual feedstuffs within a defined market area. The cost of a unit of a given nutrient, e.g., NE_L or RUP, is also estimated.

Using this method, we can compare the nutrient costs prevailing in 2004 when grains and protein prices were relatively in equilibrium to those prevailing during January 2008 after the onset of the corn ethanol euphoria (Table 3). Costs reported here are for central Ohio, where the authors have access to accurate feed markets information. These costs, as well as feed availability, would be different in the High Plains feed markets; but the analysis would be identical, yielding similar results. The cost per Mcal of NE_L in the eastern Midwest has increased by nearly 60 % during the 3-yr period, going from $0.087 to nearly $0.136/Mcal. To put this substantial increase in an even better perspective, unit cost of NE_L averaged $0.07/Mcal during the 15-yr period that extended from 1982 to 1997 and never even exceeded $0.10/Mcal during this same 15-yr period (St-Pierre and Thraen, 1999). Thus, the current price for dietary energy resides in a completely unchartered territory. Meanwhile, the cost/unit of rumen degradable protein (**RDP**) dropped by $0.12/lb between summer 2004 and winter 2008. During that same period, digestible rumen undegradable protein (**d-RUP**), non-effective NDF (**ne-NDF**), and effective NDF (**e-NDF**) unit costs increased by $0.04, $0.02, and $0.07/lb respectively. Thus, although the price of soybean meal appears to be high from a historical perspective, this doesn't imply that the prices of protein fractions (RDP and d-RUP) are also high. In fact, once we account for the increased economic value of the dietary energy contained in the protein feeds, many of them, such as expeller soybean meal are currently under-priced (Table 4). The comparison of the economic value of different commodities being traded in the Midwest yields some interesting and even surprising results. Most people would think that $4.65/bu corn implies that corn is relatively expensive. It may be so from a historical standpoint, but corn is currently a bargain feed compared to all other commodities (Table 4). What is easily overlooked is the drastic rise in the price of many other *energy* feeds in the last 3 yr, such as tallow which is now trading at nearly twice its 2004 price. During this period, however, some feeds, mostly corn

Table 3. Estimates of nutrient unit costs, central Ohio, for July 2004 and January 2008.

Nutrients	July 2004	January 2008
	---------- $/unit[1]	----------
Net energy lactation	$ 0.087	$ 0.136
Rumen degradable protein	$ 0.023	$ -0.093
Digestible-rumen undegradable protein	$ 0.342	$ 0.383
Non-effective NDF	$ -0.058	$ -0.037
Effective NDF	$ 0.054	$0.121

[1] Units are Mcal for NE_L, and lb for all other nutrients.

by-product feeds, have remained or have become extremely well priced.

MAKING BETTER USE OF BY-PRODUCT FEEDS

Strategically, one should benefit from maximizing the use of feeds deemed bargains in our prior analysis. As an example of how this can be accomplished, we balanced a ration for a 1400 lb cow producing 80 lb of milk/d at 3.7 % fat and 3.0 % protein using prices that prevailed in central Ohio during summer 2004. In doing so, we did not use a least-cost programming algorithm, but used ingredients that have traditionally formed the basis of a traditional Midwestern diet (Table 5). Using the information from Table 4, we modified the selection of ingredients to reflect the market conditions in January 2008. In doing so, we reduced the amount of corn fed by 25 % and whole cottonseed by 50 %, eliminated wet brewers grains, 44 % solvent extracted

Table 4. Actual and break-even feed prices for Central Ohio, July 2004 vs. January 2008, using Sesame III. Break-even prices were based on net energy lactation (NE_L), rumen degradable protein (RDP), digestible rumen undegradable protein (d-RUP), effective NDF (e-NDF), and non-effective NDF (ne-NDF).

	2004		2008	
Feedstuffs	Actual	Break-even	Actual	Break-even
	----------	$/ton	----------	
Alfalfa hay, 44 % NDF, 20 % CP	130	150	180	209
Bakery by-product meal	127	154	210	225
Beet pulp, dried	160	122	290	210
Blood meal, ring dried	600	540	670	632
Brewers grains, wet	35	36	35	51
Canola meal	189	181	294	224
Citrus pulp, dried	126	122	236	196
Corn grain, ground	110	156	177	231
Corn silage, 35 % DM	40	53	55	88
Cottonseed, whole w lint	208	221	319	325
Distillers dried grains w solubles	156	184	176	257
Feathers, hydrolyzed meal	330	422	385	472
Gluten feed, dry	102	155	177	217
Gluten meal, dry	369	430	559	526
Hominy	132	137	155	206
Meat meal	280	317	355	360
Molasses	105	107	160	155
Soybean hulls	112	74	178	143
Soybean meal, expeller	380	344	405	443
Soybean meal, solvent 44 %	336	253	366	290
Soybean meal, solvent 48 %	345	294	375	338
Soybean seeds, roasted	380	318	448	410
Tallow	370	358	570	558
Wheat bran	79	93	152	146
Wheat middlings	71	112	145	166

soybean meal, and tallow, and incorporated some distillers dried grains with solubles, corn hominy, corn gluten feed, and wheat middlings (Table 5, 2008). The resulting diet is nutritionally nearly identical to the traditional Midwest dairy diet. Its cost, however, is $0.25/cow/d less, resulting in estimated savings in feed costs of more than $80/lactation. Although by-products are nutritionally much more variables than grains and oilseed meals (NRC, 2001), their contribution to the nutritional variance of the whole diet is approximately proportional to the square of their inclusion rates (St-Pierre and Harvey, 1986). Because the '2008' diet uses relatively small amounts of each of the by-products, the resulting diet in fact has a lower expected nutritional variance for all major nutrients than the traditional diet (St-Pierre and Weiss, 2006).

Table 5. Comparison of a dairy diet optimized for prevailing feed prices in summer 2004 vs. January 2008.[a, b]

Ingredients	**2004**	**2008**
	------- lb as fed/d -------	
Legume hay	4.2	4.2
Legume silage	19.5	19.5
Corn silage	37.0	37.0
Wet brewers grains	13.8	0.0
Cottonseed, whole	5.0	2.75
Corn grain	15.0	11.25
Soybean meal, solvent, 44 %	2.25	0.0
Soybean meal, expeller	2.25	2.75
Dried distillers grains with solubles	0.0	3.33
Hominy	0.0	2.25
Corn gluten feed	0.0	2.75
Wheat middlings	0.0	2.75
Tallow	0.5	0.0
Minerals and vitamins	1.5	1.5
Composition		
Dry matter (lb)	51.3	51.3
NE_L (Mcal/lb)	0.74	0.73
	-------- % of DM --------	
CP	17.0	16.7
RDP	11.3	10.8
RUP	5.6	5.8
MP	10.6	10.6
NDF	32.2	33.3
NFC	42.5	42.8
Ether extracts	5.7	4.6
Cost in January 2008 ($/cow/d)[b]	4.75	4.50

[a] Rations balanced for a 1400 lb cow producing 80 lb of milk/d at 3.7 % fat and 3.0 % true protein.
[b] Prices used are those reported in Table 4 for central Ohio in January 2008.

FEEDING STRATEGIES TO OPTIMIZE PROFITS WITH HIGH FEED PRICES

Other strategies to profit optimization will be briefly discussed: 1) altering nutrient specification of diets based on feeds and milk prices, 2) increasing forage inclusion, and 3) increasing forage quality. These 3 approaches may be best used in combination.

Dairy producers should always target less than maximum milk yield in order to optimize income over feed costs and profits (St-Pierre, 1998). When milk price is relatively high and feed costs are relatively low, such as in the late 1990's, the optimum nutritional levels (i.e., nutrient concentrations in the diet) are very near those required for maximum milk production. But when milk price is low or feed prices are high, the optimal levels can be significantly less than those required to support maximum milk production. The exact mathematical optimization requires a known response function to nutritional inputs. Work has been done in this area (Bath and Bennett, 1980; St-Pierre and Thraen, 1999) but the suggested approaches lack the required precision to make them useful in practice. For the time being, different strategies can be evaluated using available ration balancing software (NRC, CPM Dairy, CNCPS, etc.) and the expert help of a professional nutritionist. Generally, forages are a cost-effective way to deliver nutrients to ruminants. Increasing inclusion of forages in place of concentrates can affect NDF, CP, RUP, NFC, and NE_L levels in a ration as well as increasing forage NDF and physically effective NDF (**peNDF**). Significant attention has been given to minimum levels of forage NDF or peNDF to ensure adequate rumination and prevent rumen acidosis and milk fat depression (Shaver, 2000; Mertens, 1994). Maximum NDF levels are determined by NE_L requirements (NRC, 2001; Mertens, 1994), and are affected by NDF digestibility.

Over the past 18 months in Ohio, corn silage, and alfalfa silage have been at break-even or lower-than-predicted values as determined using Sesame III. However, many Ohio dairy producers were not maximizing forage in rations until recently, with forages included only at 40 to 45 % of DMI. As grain prices have increased since the 2006 harvest, significant increases in forage utilization have occurred, often with little or no decrease in milk yield.

Better quality forages can be utilized more extensively in dairy rations than poorer ones. Quality is defined as the ability of the forage to deliver digestible nutrients (energy, protein, etc.) to the cow. While *in vitro* and *in situ* NDF digestibility assays are available, the poor correlation between these measures and *in vivo* NDF digestibility limits their value in predicting energy availability and lactation performance. Maturity and DM content at harvest have large impacts on forage quality, as do harvesting and storage procedures. Grain content of the small grain silages and corn silage also has a large impact on forage quality and the potential to increase forage proportions in a ration (Table 6) as well as the economic return from milk production.

TAKE-HOME MESSAGE

Many dairy producers have reduced corn inclusion in rations for lactating dairy cows by 25 to 35 % by increasing forage and by-product utilization without sacrificing milk yield or milk components. The keys to success are:

Table 6. NDF digestibility and starch content of corn silage is important in determining feeding value.[a]

Quality	Moisture (%)	NDF (%)	CWD[b] (%)	Starch (%)	Net Energy Lactation (Mcal/lb)	Partial Milk, (lb/cow)[c]	Partial Milk Income ($/cow)[d]	IOFC per 1000 cows ($/d)[e]
Poor	69.3	53.5	42.4	15.5	.453	13.1	$2.62	$1870
Fair	69.1	46.4	48.0	25.5	.526	15.3	$3.06	$2310
Medium	67.3	41.9	51.0	30.9	.561	17.3	$3.46	$2710
Good	63.3	39.7	53.8	35.2	.590	20.4	$4.08	$3330
Average	68.7	45.4	48.7	26.7	.533	15.7	$3.14	$2390

[a] Data on more than 700 samples from California kindly provided by Agri-King, Inc.
[b] CWD = Cell wall (NDF) digestibility.
[c] Predicted milk yield (lb/cow/d) from 30 lb of corn silage based on forage energy content and milk fat at 3.5 %.
[d] Milk income ($/cow/d) with milk price at $20.00/cwt.
[e] IOFC = Partial daily income over feed cost/1000 cows with corn silage priced at $50/T.

1. Maximize forage quality and optimize forage allocation.
2. Periodically question the need of every feed ingredient in your dairy rations. Cows do not have requirements for feeds (i.e., corn) but for nutrients.
3. Question narrow nutrient constraints when formulating dairy rations. Many such constraints are not well supported by *in vivo* data (i.e., cows can do quite well with diets that do not contain 25 % starch).
4. Alter diet specifications based on price of milk and feed inputs.
5. Maximize the use of bargain feedstuffs; minimize the use of over-priced feedstuffs.

LITERATURE CITED

Bath, D. L., and L. F. Bennett. 1980. Development of a dairy feeding model for maximizing income above feed cost with access by remote computer terminals. J. Dairy Sci. 63:1379-1389.

Beckman, J. L., and W. P. Weiss. 2005. Nutrient digestibility of diets with different fiber to starch ratios when fed to lactating dairy cows. J. Dairy Sci. 88:1015-1023.

Bilodeau, P. P., D. Petitclerc, N. St-Pierre, G. Pelletier, and G. J. St-Laurent. 1989. Effect of photoperiod and pair-feeding on lactation of cows fed corn or barley grain in total mixed rations. J. Dairy Sci. 72:2999-3005.

Boddugari, K., R. J. Grant, R. Stock, and M. Lewis. 2001. Maximal replacement of forage and concentrate with a new wet corn milling product for lactating dairy cows. J. Dairy Sci. 84:873-884.

Clark, P. W., and L. E. Armentano. 1997. Influence of particle size on the effectiveness of beet pulp fiber. J. Dairy Sci. 80:898-904.

DePeters, E. J., and S. J. Taylor. 1985. Effects of feeding corn or barley on composition of milk and diet digestibility. J. Dairy Sci. 68:2027-2032.

Emanuele, S. 2005. Applying starch and sugar analyses in dairy nutrition. J. Dairy Sci. 88(Suppl. 1):347.

Grant, R. 2005. Optimizing starch concentrations in dairy ration. Proceedings of the Tri-State Dairy Nutrition Conference, pp. 73-79.

Howard, W. T., and R .S. Shaver. 1997. FeedVal4 http://www.wisc.edu/dysci/uwex/nutritn/spreadsheets/FEEDVAL-Comparative.xls.

Ipharraguerre, I. R., R. R. Ipharraguerre, and J. H. Clark. 2002. Performance of lactating dairy cows fed varying amounts of soyhull pellets as a replacement for corn grain. J. Dairy Sci. 85:2905-2912.

Ipharraguerre, I. R., and J. H. Clark. 2003. Soyhulls as an alternative feed for lactating dairy cows. J. Dairy Sci. 86:1052-1073.

Ipharraguerre, I. R., J. H. Clark, and D. E. Freeman. 2005. Varying protein and starch in the diet of dairy cows. I. Effects on ruminal fermentation and intestinal supply of nutrients. J. Dairy Sci. 88:2537-2555.

Mertens, D. R. 1994. Regulation of feed intake. Pp. 450-493 in Forage Quality, Evaluation, and Utilization. J. G. C. Fahey, ed. Amer. Soc. Agronomy, Madison, WI.

Morrison, F. B. 1956. Feeds and Feeding. 22nd ed. The Morrison Publishing Company, Ithaca, NY.

National Research Council. 2001. Nutrient Requirements of Dairy Cattle, 7th revised edition. National Academy Press, Washington, D.C.

Petersen, J. 1932. A formula for evaluating feeds on the basis of digestible nutrients. J. Dairy Sci. 15:293-297.

Shaver, R. D. 2000. Feed delivery and bunk management aspects of laminitis in dairy herds fed total mixed rations. Pages 70-77 *in* Proc. III Int. Conf. on Bovine Lameness. Parma, Italy.

St-Pierre, N. R. 1998. Formulating ration based on changes in markets. Proc.Tri-State Dairy Nutrition Conference, pp. 181-203.

St-Pierre, N. R., and D. Glamocic. 2000. Estimating unit costs of nutrients from market prices of feedstuffs. J. Dairy Sci. 83:1402-1411.

St-Pierre, N. R., and W. R. Harvey. 1986. Incorporation of uncertainty of feeds into least-cost ration models. 2. Joint-chance constrained programming. J. Dairy Sci. 69:3063-3073.

St-Pierre, N. R., and C. S. Thraen. 1999. Animal grouping strategies, sources of variation, and economic factors affecting nutrient balance on dairy farms. J. Dairy Sci. 82 (Suppl2):72-83.

St-Pierre, N. R., and W. P. Weiss. 2006. *Invited.* Managing feedstuff variation in nutritional practice. J. Dairy Sci. 89 (Suppl. 1):383.

Voelker, J. A., and M. S. Allen. 2003a. Pelleted beet pulp substituted for high-moisture corn: 1. Effects on feed intake, chewing behavior, and milk production of lactating dairy cows. J. Dairy Sci. 86:3542-3552.

Voelker, J. A., and M. S. Allen. 2003b. Pelleted beet pulp substituted for high-moisture corn: 3. Effects on ruminal fermentation, pH, and microbial protein efficiency in lactating dairy cows. J. Dairy Sci. 86:3562-3570.

HOW DO FORAGE QUALITY MEASUREMENTS TRANSLATE TO VALUE TO THE DAIRY FARMER?

Normand St-Pierre and William P. Weiss[1]

ABSTRACT

The economic value of a feed resides in its content of nutrients. In dairy, the economically important nutrients are net energy lactation (**NE_L**), metabolizable protein (**MP**), effective NDF (**eNDF**), and non-effective NDF (**neNDF)**. Unit prices of the important nutrients can be calculated using the composition and prices of all feeds being traded in a given market (i.e., using a method called hedonic pricing). This method is available in a Windows-based software. Alternatively, estimates of unit prices are being published on a monthly basis in a national dairy magazine for all major dairy regions. Hay composition for NE_L, MP, eNDF and neNDF can easily be determined using 11 compositional inputs and equations provided in the appendix of this paper (largely derived from NRC, 2001). Using data from our experimental research station, we do not find a strong association between in vivo total tract NDF digestibility and in vitro NDF digestibility (NDFd). At this point we are still recommending the use of protein-free NDF and lignin to estimate NDF digestibility. For most feeds, the economic value is calculated as the simple sum of the values of their nutrients. With forages, however, one must introduce a correction associated with quality because forages are not entirely substitutable. Dairy cows exhibit a small, but significant response in milk yield when high quality forages are substituted for low quality forage in otherwise equally balanced diets. The values of three levels of quality within alfalfa and grass hay are calculated over the period of January through October 2011. In alfalfa, nearly 65% of the total value is associated with the NE_L content, whereas in grass NE_L content accounts for nearly 75% of hay values. Of the 11 compositional inputs required in the calculation of hay values, NDF, lignin and ash appears to have the greatest importance.

Key Words: alfalfa, grass, hay quality, dairy, economic value

INTRODUCTION

What are feeds used for? Animals do not require feeds; animals require nutrients. Feeds are nothing else than containers, packages of nutrients. The sole value of a feed is in the value of the nutrients that it contains. A feed containing no nutrient has no economic value. No economic value means that it is worthless - ZERO.

What are the nutrients of economic value? The answer to that question depends on two things. First, the nutrients of economic value are dependent on the class of animals under consideration. For example, the nutrients of economic importance are not the same for beef cattle, lactating

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dairy cows, dry cows, and replacement animals. Hence a given feed has a different economic value to lactating dairy cows and replacement heifers.

Second, the nutrients of economic value depend whether one is interested in the strategic value of a feed versus its tactical value. This begs further explanations. A dairy producer feeding 50 lbs/day of a finely chopped corn silage is looking for attributes in purchased hay that are very specific to the narrow conditions in which it is to be fed. The value of a given lot of hay to this producer would be entirely tactical – i.e., determined by how well it fits as a complement to other feed ingredients that are essentially pre-determined. On the other hand, a dairy producer who considers all feed components of his dairy rations to be exchangeable (tradable) would look at a given lot of hay with a strategic view. The hay would no longer be looked at as a complement to other pre-determined feeds, but as a component of the whole diet. To put it differently, tactical is when you have painted yourself in a corner; strategic is when you look at the floor configuration before you start painting.

The economic values calculated in this paper are (1) exclusively for lactating dairy cows, and (2) entirely strategic.

Of the large set of nutrients required by dairy cows, some have large economic values while others have small economic values. The calcium content of feeds is a good example of a nutrient with a small economic value. Calcium can be supplemented very inexpensively in any dairy diets. This is not to say that calcium is not important to dairy cows or that ration balancing should ignore calcium. It just says that the economics of feeding cows have little to do with calcium.

We have extensively studied the major dairy feed markets in the U.S. over a period of 30 years. Across all 3 major markets (Midwest, Northeast, West), two sets of nutrients explain over 98% of the variation in feed prices. These are:

1. Net energy for lactation (NE_L), rumen degradable protein (RDP), digestible rumen undegradable protein (dRUP), effective neutral detergent fiber (eNDF), and non-effective neutral detergent fiber (neNDF), or
2. NE_L, metabolizable protein (MP), eNDF, and neNDF.

The two sets are entirely interchangeable and give very similar results. The protein requirements of dairy cows, however, are best expressed as metabolizable protein. Therefore, nutrient set #2 will be exclusively used in the balance of this paper.

THE UNIT PRICES OF IMPORTANT NUTRIENTS

Calculating nutrient unit prices. Feed markets do exist; there are people selling and buying feeds in all major dairy regions. But there is no market for nutrients. Or is there? Can we calculate the implicit nutrient prices from the market prices of feedstuffs?

The problem of determining the implicit price of attributes (the nutrients) embedded in various products (feedstuffs) is no at all unique to the feed/nutrient complex. Economists have found an elegant way to solve this problem using a method called hedonic pricing. We will not review the details of how this work in this paper. Interested reader can consult St-Pierre and Cobanov

(2000) for further details. In short, prices and nutritional composition of all ingredients traded in a given market are used to back-calculate using statistical methods what the markets are implicitly pricing the nutrients contained in feeds.

Market prices of nutrients in the Midwest for the 80 months between January 2005 and September 2011 are shown in Figure 1. During this period, the cost per unit of NE_L has more than tripled, the cost per unit of MP has doubled, while the cost per unit of eNDF has seen a drastic surge over the last 12 months. The U.S. has experienced drastic changes in renewable energy policies during the last decade, some of which have had a substantial effect on feed prices. However, large variation in nutrient unit prices are still evident even over a much shorter period of time such as what has occurred since January 2011 (Table 1).

What affects nutrient unit prices? Market prices of all feeds affect nutrient unit prices, not just the prices of corn and soybean meal. Therefore, nutrient unit prices change through time and location. We have already shown how nutrients can quickly change through time in Table 1. Because of regional differences in feed prices and availability, nutrient unit prices also show substantial regional differences (Table 3). In this paper we will be using Midwest nutrient unit prices. When examining the effect of time, we will use January to October 2011 prices.

Where do I find the nutrient unit prices? You can purchase a Windows-based software that we wrote (*Sesame*) for $10 at www.sesamesoft.com. Beware that it is NOT the easiest software to use. Alternatively, we publish a regular column in *Progressive Dairyman* where we publish the nutrient unit prices for the major dairy regions of the U.S.

FORAGE COMPOSITION

Nutrient composition used in this paper. Each lot of forage has a unique nutrient composition that affects its value. The nutrient composition of forages used as examples in this paper are reported in Table 2. We used 3 levels of quality for legume hay and grass hay.

A few things are worth mentioning here. First, notice that the quality of alfalfa (and grass) has a much smaller effect on its metabolizable protein than on its crude protein. Metabolizable protein is the sum of the digestible rumen undegradable protein (dRUP) and the digestible microbial true protein (dMTP). The protein of forages is largely degraded in the rumen (70 to 75%), thus making little contribution to dRUP. Microbial protein synthesis is entirely driven by the digestibility of the nutrients (i.e., the TDN content) in the NRC (2001) model. The level of intake of an animal affects TDN: the TDN of a feed for a cow at maintenance is greater than its TDN for a cow producing milk. For lactating cows, at least for now, TDN and NE_L should be reported at 3 times maintenance (i.e., for milk production of 60 to 70 lbs/day).

Second, observe that in alfalfa the non-fiber carbohydrates (NFC) contribute 2 times more to its energy content (i.e., TDN) than the NDF. In grass, the situation is reversed, with NDF contributing significantly more to the energy content than NFC. Total tract NDF digestibility is greater in grass (~47%) than in alfalfa (~39%) of equivalent quality.

How are NE_L and MP calculated? Equations used in the calculation of NE_L and MP according to NRC (2001) are reported in the appendix. Although these equations may seem intimidating at first, they can easily be programmed in a computer spreadsheet.

While 9 chemical entries are required for the calculation of NE_L only 5 measurements will have much effect on NE_L in practice: DM, CP, NDF, lignin, and ash. Other entries can simply be found in feed composition tables.

The calculation of MP requires a measurement of CP, rumen degradability of protein (RUP_CP), and post-ruminal digestibility of RUP (RUPd). Some feed laboratories provide estimates of RUP_CP, but the variation within a type of hay is relatively small and largely inconsequential to the value of the forage. Likewise, table values for RUPd should be used.

What NDF digestibility should be used? The ratio of lignin to protein-free NDF is used to estimate in vivo total tract NDF digestibility (TT-NDFd) in the equation for TDN_NDF reported in the Appendix. The 2/3 exponent is used to convert mass to surface area, hence representing the decrease in NDF digestibility due to the surface interaction (i.e., coating) of cell walls by lignin. This conceptual interaction is necessarily a simplification of the complex anatomy and chemistry of plant cell walls. Some have advocated the use of in vitro NDF digestibility (NDFd) as a proxy for the calculated TT-NDFd used when calculating TDN_NDF. Although this approach is appealing, much doubt remains regarding the relationship between NDFd and TT-NDFd. For example, we have summarized the relationship between TT-NDFd and NDFd for 23 diets where TT-NDFd was measured using total fecal collection in trials conducted at our experimental research station (Figure 2). In this figure, it is apparent that NDFd overestimates differences between treatments and that the magnitude of the difference in NDFd is not related to the magnitude of the difference in TT-NDFd. The ranking within experiment was often OK with NDFd, raising the possibility of using NDFd for energy calculation. It is clear, however, that NDFd cannot be directly substituted for TT-NDFd when calculating the energy of a feed. Much work is needed in this area. Meanwhile, we still recommend using the equation with the ratio of lignin to protein-free NDF for estimating the energy contribution of NDF.

CALCULATING THE VALUE OF A FORAGE

The value of the nutrients. So far, we have shown that nutrient unit prices can be calculated from market information from all feedstuffs traded in an area. We also explained how the nutrient composition of forages for the economically important nutrients are calculated. Determining the value of the nutrients in a given forage involves a series of simple arithmetical operations. These are illustrated for the reference alfalfa in Table 4 using the average price of nutrients for the Midwest in 2011.

Correcting for milk production response. For most feeds, the sum of the values of the nutrients as calculated in the preceding paragraph is its average economic value; but not for forages. While most feeds are substitutable based on their nutrient content, this is not entirely true for forages. What this means is that two rations balanced for exactly the same nutrient density (NE_L, MP, eNDF, neNDF) but using forages of different quality do not result in exactly the same milk production. Cows fed the ration based on a high quality forage respond to forage quality with additional milk production mainly through greater dry matter intake (DMI). Note that this is not the same as the response to feeding forages of different quality, but without any ration re-balancing. Here the rations are identical in their nutritional content, but cows fed rations based on higher quality forages achieve a greater level of milk production.

We used results from many research trials to calculate the response to forage quality. Although one could think of a better marker of quality than the total NDF content of forage, the data did not allow the calculation of anything more than NDF. The resulting equations are shown in Figures 3 and 4 for alfalfa and grass, respectively. Algebraically, the equations to calculate the value per ton of forage due to milk production responses are:

Alfalfa: Value of Response ($/ton) = [(P-Milk x 0.273 x (44 – NDF)] x DM ÷ 100

Grass: Value of Response ($/ton) = [(P-Milk x 0.3 x (53 – NDF)] x DM ÷ 100

where P-Milk is the price of milk ($/cwt). It is important to understand that this adjustment to the value of forages means that forage values are dependent on milk prices. The difference in the value of a high quality forage compared to that of a low quality forage is much smaller when milk prices are low (such as in 2009) then when milk prices are high (as in 2011).

Effect of forage type and quality on the value of forage in 2011. Table 5 summarizes the values of alfalfa and grass hay for 3 levels of quality for the period of January to October 2011. On an average, the range in values due to quality is greater in grass ($124/ton) compared to alfalfa ($73/ton). This could be due to the arbitrary range of quality selected for the two types of forages. On an average, alfalfa is worth $50/ton more than grass. Compared to alfalfa, grass hay shows a greater range in value through time: $109 and $126/ton for alfalfa and grass, respectively. More importantly, the value of forages changes considerably through time even over a short time span of only 10 months.

Contribution to calculated values. The value of a forage is the sum of the values of its important nutrients plus the milk response associated with the forage quality expressed as NDF content. The contribution of each nutrient to the value of a forage is not the same. Table 6 shows the average contribution of NE_L, MP, eNDF, neNDF, and milk response for the first 10 months of 2011 for the reference alfalfa and grass hays. On an average, energy (NE_L) content accounts for nearly 65% of the value of alfalfa hay and 75% of the value of grass hay. Notice that the protein in alfalfa (20% crude protein) is only worth $7/ton more than the protein in grass (12% crude protein). This is because most of the protein in forage is rumen degradable (RDP), and the unit value of RDP ($/lb) is generally null and often even slightly negative (results not shown). Producing forages of greater protein content is of little value unless the protein increase is associated with an increase in the digestibility (energy) of the resulting feeds.

In long hay, most of the NDF is effective, resulting in very small neNDF content (Table 2). Therefore, the contribution of neNDF to the value of long hays could be entirely ignored without meaningful losses in the accuracy of estimating their economic values.

Marginal changes from compositional values. The 4 nutrients used to calculate the value of a forage are calculated from 11 compositional entries. We can calculate the change in the value of our reference alfalfa and grass hay from a one-unit change in each of the 11 entries (Table 7). One should be extremely careful in the interpretation of these results. First, a one-unit change does not represent the same degree of "difficulty" across all compositional elements. For example, it is considerably easier to raise the NDF of grass by one unit than to raise its ether extracts also by one unit. Second, and even more importantly, it is very difficult in nature to change a compositional element by one unit without affecting any of the other compositional

elements. For example, crude protein in alfalfa is negatively associated with NDF content. On average, raising crude protein lowers the NDF content (i.e., the plant is more immature). Likewise, NDF and lignin content are positively associated, meaning that an increase in NDF content is generally associated with an increase in lignin content. Keeping these reservations in mind, compositional elements can be loosely grouped into 3 categories.

1. Those with a small effect on the value of hay: RUP, RUPd and NDFe,
2. Those with a medium effect on the value of hay: DM, CP, NDICP, and
3. Those with a large effect on the value of hay: ADICP, ether extracts, NDF, lignin and ash.

Ether extracts and ADICP are relatively constant within hay type compared to NDF, lignin and ash. NRC (2001) reports standard deviations of 0.4 (ADICP), 0.5 (ether extracts), 0.9 (lignin), 1.0 (ash), and 6.3 (NDF) percent. Thus, NDF, lignin, and ash are arguably the most influential compositional entries to hay values. Note that the net effect of NDF on hay values incorporates its positive effect on forage value from its positive contribution to eNDF, and its negative effects on forage value from its negative contribution to NE_L and MP.

CONCLUDING REMARKS

The values calculated in this paper are on a farm-gate basis (i.e., delivered) and not FOB. Therefore, forage growers would have to account for delivery costs when estimating the value of a given hay. In addition, the values calculated in this paper are averages and represent what coherent buyers should be willing to pay. Coherent behavior is often an elusive attribute in feed markets. Some buyers consistently shop for “supreme” quality alfalfa hay because that’s what they have been using for years without ever considering whether other quality levels or even other types of feeds make more economic sense. A very thirsty man is much more willing to pay an exorbitant price for a cold beer, especially if he fails to consider the free water available from a fountain nearby.

REFERENCES

St-Pierre, N. R., and D. Glamocic. 2000. Estimating unit costs of nutrients from market prices of feedstuffs. J. Dairy Sci. 83:1402-1411.

APPENDIX

Equations used to calculate NE_L (NE_3X, Mcal/lb) of forages:

TDN_NFC = 0.98 x (100 – NDF + NDICP – CP – EE – ASH)

TDN_NDF = 0.75 x (NDF – NDICP – LIG) x $(1-(LIG/(NDF-NDICP))^{0.667})$

TDN_CP = CP x exp(-1.2 x ADICP/CP)

TDN_EE = (EE – 1) x 2.25

TDN_1X = TDN_NFC + TDN_NDF + TDN_CP + TDN_EE - 7

DE_1X = (TDN_NFC x 0.042) + (TDN_NDF x 0.042) + (TDN_CP x 0.056) + ((EE-1) x 0.094) - 0.3

TDN_3X = TDN_1X x 0.92

NE_3X = (0.6532 x DE_1X) - 0.5064

Equations used to calculate the metabolizable protein (MP, % of DM) of forages:

dRUP = CP x RUP_CP x RUPd ÷ 10000

dMTP = TDN_3X x 1.3 x 0.64 ÷ 10

MP = dRUP + dMTP

Equations used to calculate effective and non-effective NDF of forages:

eNDF = NDF x NDFe

neNDF = NDF - eNDF

where:

ADICP	=	ADF insoluble crude protein (% of DM)
ASH	=	Ash (% of DM)
CP	=	Crude protein (% of DM)
dRUP	=	Digestible RUP (% of DM)
dMTP	=	Digestible microbial true protein (% of DM)
EE	=	Ether extracts (% of DM)
eNDF	=	Effective NDF (% of DM)
exp	=	The exponential function (i.e., *e* exponent the value in parentheses)
LIG	=	Lignin (% of DM)
NDF	=	Neutral detergent fiber (% of DM)
NDFe	=	NDF effectiveness (% of NDF)
NDICP	=	NDF insoluble crude protein (% of DM)
NE_3X	=	Net energy for lactation measured a 3 times maintenance (Mcal/lb)

neNDF	=	Non-effective NDF (% of DM)
RUP_CP	=	Rumen undegradable protein (% of CP)
RUPd	=	Digestibility of RUP (% of RUP)
TDN_1X	=	Total digestible nutrients at 1 time maintenance (% of DM)
TDN_3X	=	Total digestible nutrients at 3 times maintenance (% of DM)
TDN_CP	=	TDN from the crude protein fraction (% of DM)
TDN_EE	=	TDN from the ether extracts fraction (% of DM)
TDN_NDF	=	TDN from the NDF fraction (% of DM)
TDN_NFC	=	TDN from the NFC fraction (% of DM).

Table 1. Nutrient unit prices in the Midwest between January and October 2011.

Nutrients	Average	S.D.	Min	Max
NE_L (¢/Mcal)	15.2	2.1	11.1	17.4
MP (¢/lb)	31.1	4.8	26.8	40.6
eNDF (¢/lb)	4.7	4.0	0.0	11.9
neNDF (¢/lb)	-8.5	2.8	-12.4	-5.1

Table 2. Nutrient composition of the reference alfalfa hay, reference grass hay, and low and high quality hays used as examples.

		Alfalfa			Grass		
Nutrients[1]	**Units**	**Reference**	**Low**	**High**	**Reference**	**Low**	**High**
Dry matter	%	88	88	88	88	88	88
Crude protein	%	20	16	24	12	8	16
NDICP	%	2.5	2.5	2.5	4.0	4.0	4.0
ADICP	%	1.5	1.5	1.5	1.0	1.0	1.0
Ether extracts	%	2.0	2.0	2.0	2.5	2.5	2.5
NDF	%	40	44	36	60	68	52
ADF	%	30	34	26	40	48	32
Lignin	%	7.0	8.8	5.4	6.5	8.5	4.5
Ash	%	10	10	10	7	7	7
RUP	%CP	25	25	25	30	30	30
RUPd	% RUP	70	70	70	65	65	65
Effective NDF	% NDF	92	92	92	98	98	98
TDN from NFC	%	29.9	29.9	29.9	22.1	18.1	26.0
TDN from NDF	%	15.4	15.8	14.8	28.3	30.8	25.9
TDN from CP	%	18.3	14.3	22.3	10.9	6.9	14.8
TDN from EE	%	2.3	2.3	2.3	3.4	3.4	3.4
TDN at 3X	%	54.0	50.7	57.1	52.9	47.9	57.9
NE_L at 3X	Mcal/cwt	57.6	51.5	63.5	53.0	44.6	61.5
MP at 3X	%	7.99	7.02	8.95	6.73	5.55	7.94
		---------- Units per Ton ----------					
NE_L	Mcal	1014.2	906.7	1118.1	932.6	785.5	1082.3
MP	lbs	140.7	123.5	157.6	118.6	97.6	139.7
eNDF	lbs	647.7	712.4	582.9	1034.9	1172.9	896.9
neNDF	lbs	56.3	62.0	50.7	21.1	23.9	18.3

[1] NDICP = NDF insoluble crude protein; ADICP = ADF insoluble crude protein; NDF = neutral detergent fiber; ADF = acid detergent fiber; RUP = rumen undegradable protein; RUPd = RUP digestibility; TDN = total digestible nutrients; NE_L at 3X = net energy for lactation calculated at an intake of 3 times maintenance; MP at 3X = metabolizable protein calculated at 3 times maintenance.

Table 3. Nutrient unit prices across 3 regions[1], November 2011.

Nutrients	Northeast	Midwest	West
NE_L (¢/Mcal)	12.7	8.6	10.0
MP (¢/lb)	32.2	34.4	42.6
eNDF (¢/lb)	6.2	11.3	15.4
neNDF (¢/lb)	-3.8	5.5	1.4

[1] Northeast prices are for NY and northern PA; Midwest prices are for WI and eastern MN; West prices are for the San Joaquin Valley of CA.

Table 4. Calculation of the value of the nutrients in one ton of the reference alfalfa hay.

	Composition	DM %	Mcal or Pounds per Ton	Unit Prices ¢	Value $/ton
NE_L (Mcal/cwt)	57.6	88	1014.2	15.2	154.15
MP (% DM)	7.99	88	140.7	31.1	43.76
eNDF (% DM)	36.8	88	647.7	4.7	30.44
neNDF (% DM)	3.2	88	56.3	-8.5	-4.79
Total					**223.56**

Table 5. Value ($/ton) of alfalfa and grass hay of 3 quality levels, Midwest, January to October 2011.

Feeds	Average	SD[1]	Min	Max
Alfalfa - Reference	241	37	193	302
- Low	204	36	162	265
- High	277	38	223	338
Grass - Reference	191	46	145	271
- Low	129	47	77	210
- High	253	47	197	332

[1] SD = standard deviation.

Table 6. Average contribution of NE_L, MP, eNDF, neNDF, and milk response to the value of our reference alfalfa and grass hays.

	Alfalfa		**Grass**	
Component	**$/ton**	**% of Total**	**$/ton**	**% of Total**
NE_L	154.55	64.0	141.75	73.7
MP	43.75	18.2	36.88	19.2
eNDF	30.44	12.6	48.64	25.3
neNDF	-4.78	-2.0	-1.79	-0.9
Milk	17.29	7.2	-33.26	-17.3
Total	240.86	100.0	192.21	100.0

Table 7. Marginal change in value of forage hay ($/ton) from a one-unit increase in each of the compositional entries.

	Change in forage value ($/ton)	
Composition entries	Alfalfa - Reference	Grass - Reference
Dry matter (%)	2.73	2.19
Crude protein (% DM)	2.12	2.23
NDICP (% DM)	1.28	1.14
ADICP (% DM)	-5.17	-5.01
Ether Extracts (% DM)	4.72	4.72
NDF (% DM)	-4.96	-5.10
Lignin (% DM)	-4.42	-5.38
Ash (% DM)	-3.68	-3.67
RUP (% CP)	0.76	0.43
RUPd (% RUP)	0.27	0.20
NDF Effectiveness (% NDF)	0.93	1.40

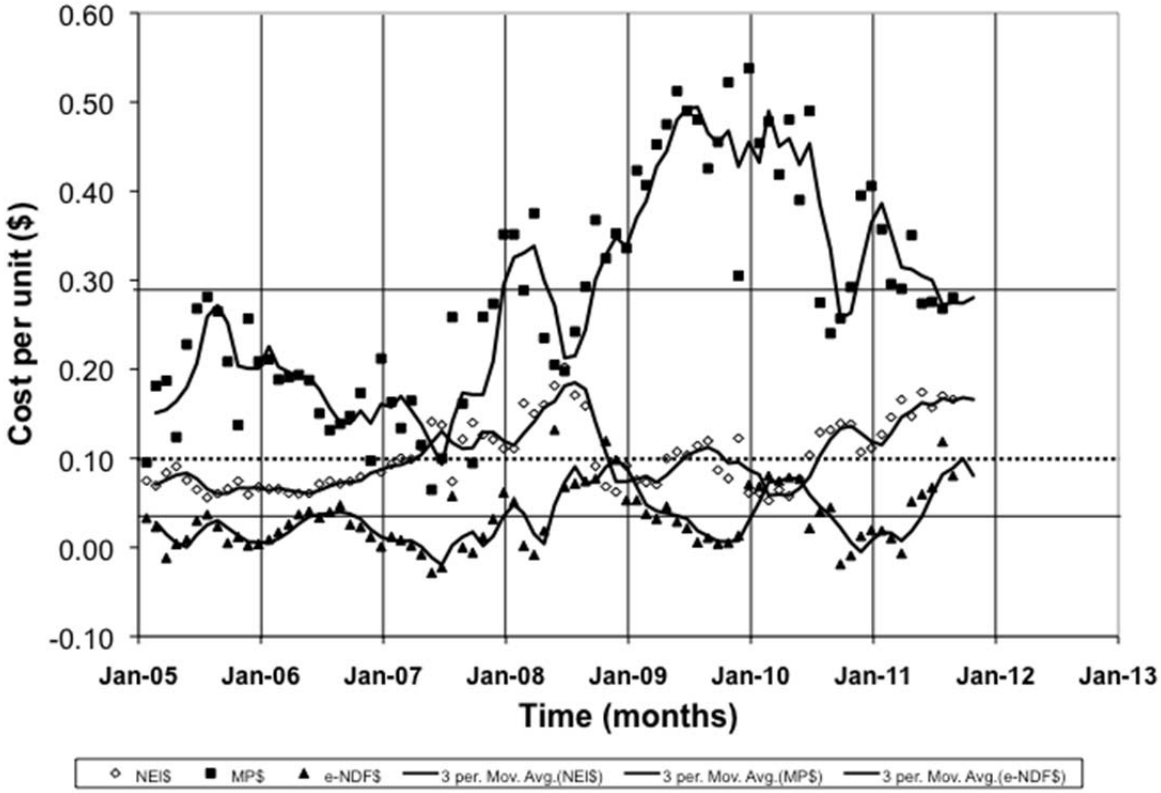

Figure 1. Price of nutrients in the Midwest from January 2005 through September 2011. NEl$ = Net energy for lactation ($/Mcal), MP$ = metabolizable protein ($/lb), and e-NDF$ = effective NDF ($/lb).

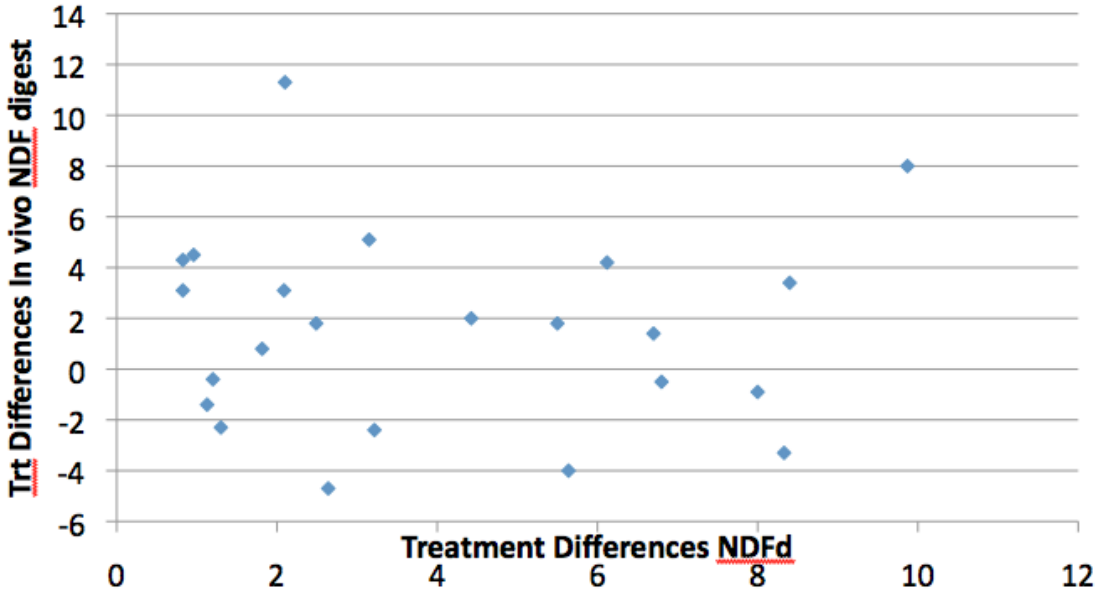

Figure 2. Relationship between whole diet total tract in vivo NDF digestibility (TT-NDFd) expressed as deviation from a control diet and in vitro NDF digestibility (NDFd).

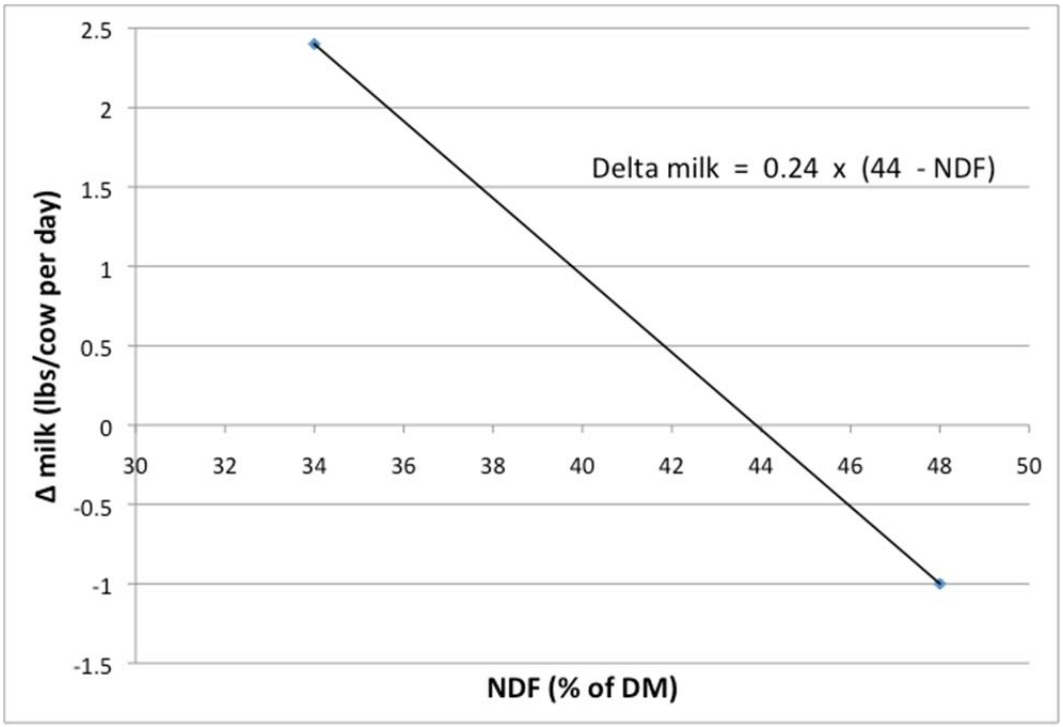

Figure 3. Response in milk production to change in NDF content of alfalfa forage.

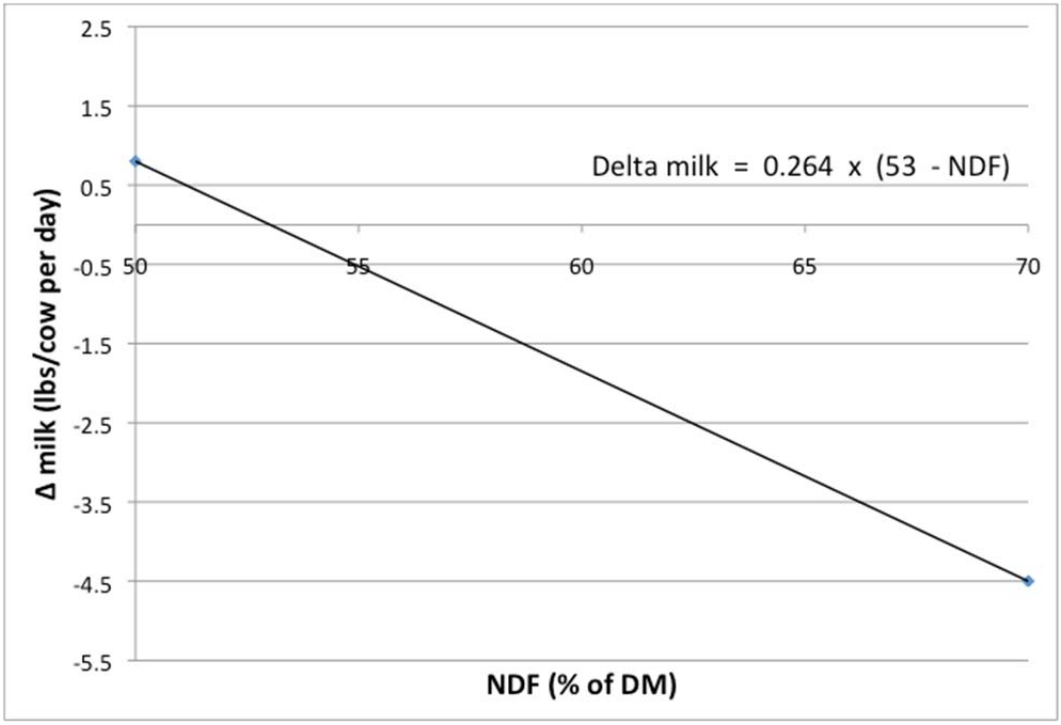

Figure 4. Response in milk production to change in NDF content of grass forage.

SHOULD FORAGES BE PRICED LIKE A SACK OF NUTRIENTS?

Bill Weiss and Alex Tebbe[1]

ABSTRACT

The economic value of forage should reflect its nutrient composition relative to market conditions. The major nutrients provided by forage are energy (expressed as net energy, NEL), protein (expressed as metabolizable protein, MP) and fiber (expressed as neutral detergent fiber, NDF) which can be assayed or calculated by commercial feed labs. By using a statistical technique and the nutrient composition and prices of multiple feeds, including forages, we can determine the average economic value of those nutrients. With those two pieces of information, we can estimate the economic value of a feed based on the average cost of nutrients within a specific market and its nutrient composition. For example alfalfa hay may have 0.62 Mcal/lb of NEL, 13%MP and 40% NDF (100% DM basis) and in central Ohio in September, 2019 average NEL was worth $0.09/Mcal, MP was worth $0.30/lb and forage NDF was worth $0.12/lb. In this example 1 ton of this alfalfa (85% DM) has 1054 Mcal of NEL worth $95, 221 lbs of MP worth $66 and 680 lbs. of NDF worth $82 for a total value of $243/ton. However, pricing totally on nutrients ignores the effect forage can have on feed intake, which ultimately affects growth or milk production. Concentration of NDF and in vitro NDF digestibility, relative to some standard can be used to adjust the value of forages but the adjustment depends on the type of forage (e.g., grass, legume, corn silage), milk price and diet cost.

Pricing Nutrients

The inherent economic value of any feed is its ability to support productive functions when fed to livestock. However, because dairy cows are fed diets, not single ingredients, we cannot describe feeds in terms of production potential. For example, we cannot say that 10 lbs. of a specific alfalfa will support 15 lbs. of milk. However, we can say, with reasonable accuracy, that 10 lbs. of a specific diet will support 15 lbs. of milk. The amount of milk or body weight gain a diet will support is a function of the nutrients in the diet, and the amount of the diet an animal can eat. Therefore, rather than valuing a forage (or any feed) based on its ability to support growth or milk, we can value a feed based on its nutrient composition. Forages are routinely sampled and analyzed for chemical components and with that information we can calculate or estimate the concentration of major nutrients. The NEL concentration of forage is mostly a function of NDF content (higher = lower NEL), NDF digestibility (higher = higher NEL), and ash (higher = lower NEL). Labs have equations to estimate NEL and that value is provided on their reports.

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Metabolizable protein is a function of the type of crude protein and energy intake and can only be estimated accurately for a diet, not a feed. Therefore, it is not usually put on lab reports. However, it can be estimated for a feed by assuming average conversion of rumen degradable protein (RDP) into microbial protein and using feed specific digestibility coefficients for rumen undegradable protein (RUP) from the NRC (2001). In this example (Table 1), we assumed the forages are part of a well formulated diet sufficient in NEL and RDP and for typical dairy cows consuming 4% of their body weight as DM. The amount of CP as RDP and RUP are from NRC (2001). Similarly, for digestible RUP, feed specific digestion coefficients are used to convert RUP into the protein available to the cow (i.e., the MP). The digestibility of RUP and the amount of RDP typically decreases as forage quality decreases. If cows are fed exactly to their RDP requirement, the efficiency of converting RDP into MP is 0.55; however because RDP is usually fed in slight excess an efficiency of 0.53 is more realistic. The total MP of the feed is then the sum of digestible RUP and RDP x 0.53. To estimate MP for a specific forage, find the forage in Table 1 that is most similar to your forage and multiply the assayed CP value by the MP efficiency (MP/CP) shown in the table.

Table 1: Example of calculating metabolizable protein for common forages and soybean meal.

		% of CP			
Ingredient	CP, %DM	RDP[1]	dRUP[2]	MP[3], %DM	MP/CP
Good alfalfa hay, 85% DM	23.0	82.3	13.3	13.1	0.57
Alfalfa hay, 85% DM	18.0	75.9	15.7	10.1	0.56
Corn silage, 35% DM	8.8	64.7	24.7	5.3	0.59
Grass hay, 85% DM	13.3	69.5	19.8	7.5	0.57
Soybean meal, 89% DM	53.8	57.4	39.6	37.7	0.70

[1] Rumen degradable protein, % CP = 100 – RUP, %CP
[2] Digestible rumen undegradable protein, % CP = RUP, %CP × RUP digestion, %
[3] Metabolizable protein = CP × (RDP × 0.53 + dRUP) ÷ 100, %DM

Labs routinely measure and report NDF concentrations of feeds. Different expression of NDF are reported (e.g., ash-free NDF or NDF_{OM}), but for this discussion any measure of NDF is adequate. The NDF in forages provides some energy but it has unique economic value because it is needed to maintain rumen health. Fiber from other feeds provides energy but have little effect on rumen health so that fiber has less value than forage fiber.

Using computer software (e.g., SESAME) and nutrient composition and ingredient prices for several feeds (including forages) within a specific market, the average economic value of the major nutrients can be estimated. In this example, we used the average nutrient values in September, 2019 for central OH (Table 2; from Tebbe, Buckeye Dairy News, September 2019). These values are specific

Table 2. Prices of dairy nutrients for OH, September 2019

Nutrient,	Estimate
NE_L, $/Mcal	0.09 ± 0.017*
MP, $/lb	0.30 ± 0.053*
e-NDF, $/lb	0.12 ± 0.039*
ne-NDF, $/lb	-0.05 ± 0.033

* estimate is not equal to zero ($P < 0.01$)

to central OH and do not reflect the prices on a historical basis. Price estimates can also have a negative value meaning they are discounted. The discount is primarily because the two main drivers of ingredient markets (i.e., chickens and ethanol) do not want fiber. This is common for non-effective NDF; however, forage fiber is almost all effective NDF and should not be discounted. Monthly nutrient prices are also available in Progressive Dairy magazine for six different regions in the U.S (https://www.progressivedairy.com/).

With the nutrient composition of a forage and average nutrient prices, we can determine the economic value of a feed within a specific market. For example, alfalfa hay has 0.62 Mcal/lb of NEL, 13%MP and 40% NDF (100% DM basis) and in central Ohio in September, 2019 average NEL was worth $0.09/Mcal, MP was worth $0.30/lb and forage NDF was worth $0.12/lb. In this example 1 ton of alfalfa (85% DM) has 1054 Mcal of NEL worth $95, 221 lbs of MP worth $66 and 680 lbs of NDF worth $82 for a total value of $243/ton.

The total value calculated ($243/ton) would be the average estimate of this alfalfa; however, estimates have inherent variation and must be considered. The inherent variation accounts for variation in the nutrient composition and market prices for ingredients used to calculate the price estimates in Table 2. The variation in nutrient price estimates can be used to calculate our confidence in the estimate. For example, with 75% confidence, the true NE_L value of premium alfalfa hay is between $74 – 115/ton [i.e., $95/ton ± (0.017 × 1.15 × 1054 Mcal of NE_L)]. With 95% confidence, the range in true NE_L value for premium alfalfa hay is wider and between $60 – 130/ton [i.e., $95/ton ± (0.017 × 1.96 × 1054 Mcal of NE_L)]. The same calculation can be applied to the other nutrients. Calculating the 75% confidence intervals of the final estimate is much more complicated and beyond the scope of this talk. A good estimate for the calculating 75% interval for forages would be ± 12.5% of the mean or $215 – 275/ton for the premium alfalfa. The take away message is that the total value calculated should have a range.

This approach works quite well when comparing corn grain to barley grain or soybean meal to canola meal because differences in nutrient composition between those feeds account for most of the differences in milk yield or growth when those feeds are fed. Factors in forages other than basic nutrients can affect feed intake which can have profound effects on animal productivity. Adjusting prices for effects on intake is difficult because the effect a forage has on intake depends on the inclusion rate of the forage, the make-up of the entire diet, and the milk yield and physiological state of the cow.

Adjusting for forage quality

Forages are now commonly analyzed for in vitro NDF digestibility (IVNDFD) which is probably the best single assay to estimate effect of forage on intake and production. Cows consume higher amounts of feed when fed diets that contain forages with increasing IVNDFD. Labs use different incubation times but 30 and 48 hour are most common, and either incubation time point can be used to adjust nutrient prices. On average a 1% unit increase in IVNDFD (expressed as % of NDF) increases intake by 0.26 lbs. and increases milk yield by 0.47 lbs./day (Oba and Allen 2005). Those values are appropriate for a change in IVNDFD, not an absolute value. For

example, if cows were changed from a diet with a forage that had an IVNDFD of 50% to one with an IVNDFD of 55%, we would expect milk to increase by 5 x 0.47 = 2.4 lbs./day. We would expect the same increase when IVNDFD increased from 35 to 40%. Because we can only evaluate change in IVNDFD, we have to compare the forage of interest to a standard (Table 3). We chose to set the standard equal to mean values for alfalfa, grass and corn silage (means obtained from DairyOne, Ithaca, NY; https://dairyone.com/services/forage-laboratory-services/feed-composition-library/).

Table 3. Average NDF and in vitro NDF digestion of common forages

		Mean IVNDFD, % of NDF	
Forage	Mean NDF, % of DM	30 hour	48 hour
Alfalfa	39	41	47
Corn silage	43	53	62
Cool season grasses	57	61	65

To calculate the quality adjustment, the difference between IVNDFD of the forage of interest and standard values are calculated (use the same incubation time period for both the sample and standard): IVNDFD(sample) – IVNDFD(standard). That value can be positive or negative. It is then multiplied by 0.47 to estimate the change in milk yield when the forage is fed. Change in milk yield then has to be converted to a dollar value, which is a function of milk price and feed price. Based on the data from Oba and Allen (2005), dry matter intake is expected to increase 0.55 lbs for every 1 lb. increase in milk yield (conversely if milk yield drops by 1 lb. we expect dry matter intake to decrease 0.55 lbs). The value of the change in dry matter intake depends on the price of the diet. Lastly, to put these numbers on a per ton of forage dry matter basis, an intake of forage dry matter must be assumed. We chose 22 lbs (55 lbs of dry matter intake that was 40% of the forage of interest).

Example calculation of quality adjustment.

The forage of interest is alfalfa hay that had a 48 hour IVNDFD of 55%.

1. Difference in IVNDFD from standard: 55 – 47 = 7 units
2. Expected increase in milk yield: 7 x 0.47 = 3.3 lbs/day (assumed milk price 0.19/lb.)
3. Expected increase in DM intake: 7 x 0.26 = 1.8 lbs (assumed feed price $0.08/lb DM)
4. Expected gain in income over feed cost: (3.3 x 0.19) – (1.8 x 0.08) = $0.49
5. Converting to 1 lb. of forage DM: 0.49/22 = $0.022/lbs = $44.5/ton of DM or about $38/ton of hay (85% DM).

That value is added (or subtracted) from the nutrient value calculated as described above. In the example above the alfalfa had a nutrient value $243/ton (85% DM) and a quality adjustment of $38/ton; therefore the total value of the example alfalfa is $281/ton. Table 4 has quality adjustments for various feed and milk prices. A user would match feed and milk price most applicable, find the quality adjustment (in $/ton of DM) and add (or subtract) it from the nutrient value.

The values calculated using this method is the absolute maximum a dairy farmer should pay because basically it represents break-even cost for the forage (in other words, the dairy producer is getting what he paid for but the forage is not a bargain or overpriced)

Table 4. Quality adjustment ($/ton of forage DM) per 1 percentage unit change in IVNDFD from the standard. For example, if the forage had 3 percentage units lower IVNDFD than the standard and milk price was $22/cwt and diet cost $0.10/lbs of DM, the quality adjustment would be -3 x 7.0 = $-21/ton.

	Milk Price, $/cwt					
Feed price, $/lbs of DM	**16**	**18**	**20**	**22**	**24**	**26**
0.06	5.4	6.3	7.1	8.0	8.6	9.7
0.08	5.0	5.8	6.7	7.5	8.4	9.2
0.10	4.5	5.3	6.2	7.0	7.9	8.8
0.12	4.0	4.9	5.7	6.6	7.4	8.3

Limitations to the method

1. We assume the effect of a change in IVNDFD is the same regardless of forage inclusion rate, which is most likely not true. At low inclusion rates we are probably overestimating the value of quality and at high rates we are likely underestimating the value. We chose an inclusion rate of 40%
2. We assumed the same effect of a change in IVNDFD for all forages. This may or may not be true. The majority of the data in the paper by Oba and Allen (2005) is from corn silage based diets, however several studies also included alfalfa. Grass was poorly represented in the data set.
3. We assumed the same effect of a change in IVNDFD for all milk yields. Most likely high producing cows will be more sensitive to a change in IVNDFD than lower producing cows.

CONCLUSIONS

Forages are more than just a sack of nutrients, however the initial value of forage should be based on the nutrients the forage prices and the value of those nutrients. Pricing forage only on an energy, protein and fiber basis ignores the effect (positive or negative) forage quality has on milk yield and income over feed costs. The use of IVNDFD can help put a dollar value on forage quality. Although our method has limitations, on average it is an acceptable way to price forage.

REFERENCES

Oba, M. and M. Allen. 2005. In vitro digestibility of forages. Pages 81-91 in Proc. Tri-State Dairy Nutrition Conf., Ft. Wayne, IN.